

ANATOMY OF THE ORAL CAVITY, PALATE & SALIVARY GLANDS

By

Dr. Faris Muhammed

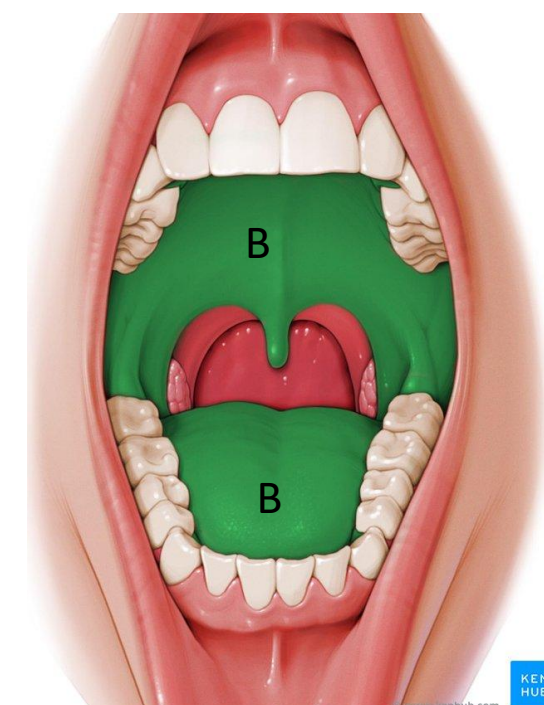
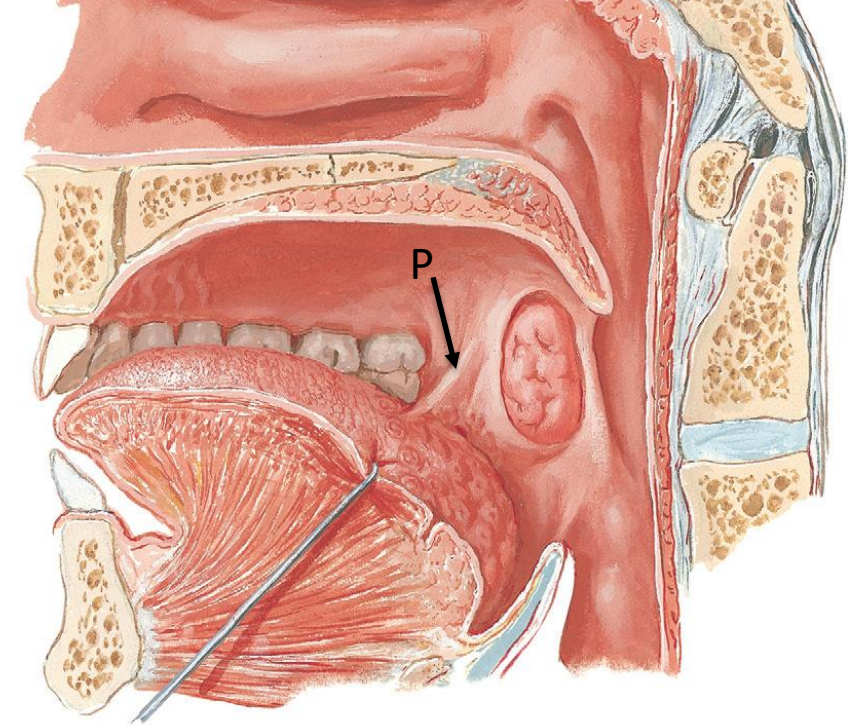
PhD clinical Anatomy

College of Medicine, HMU

6.9.2026

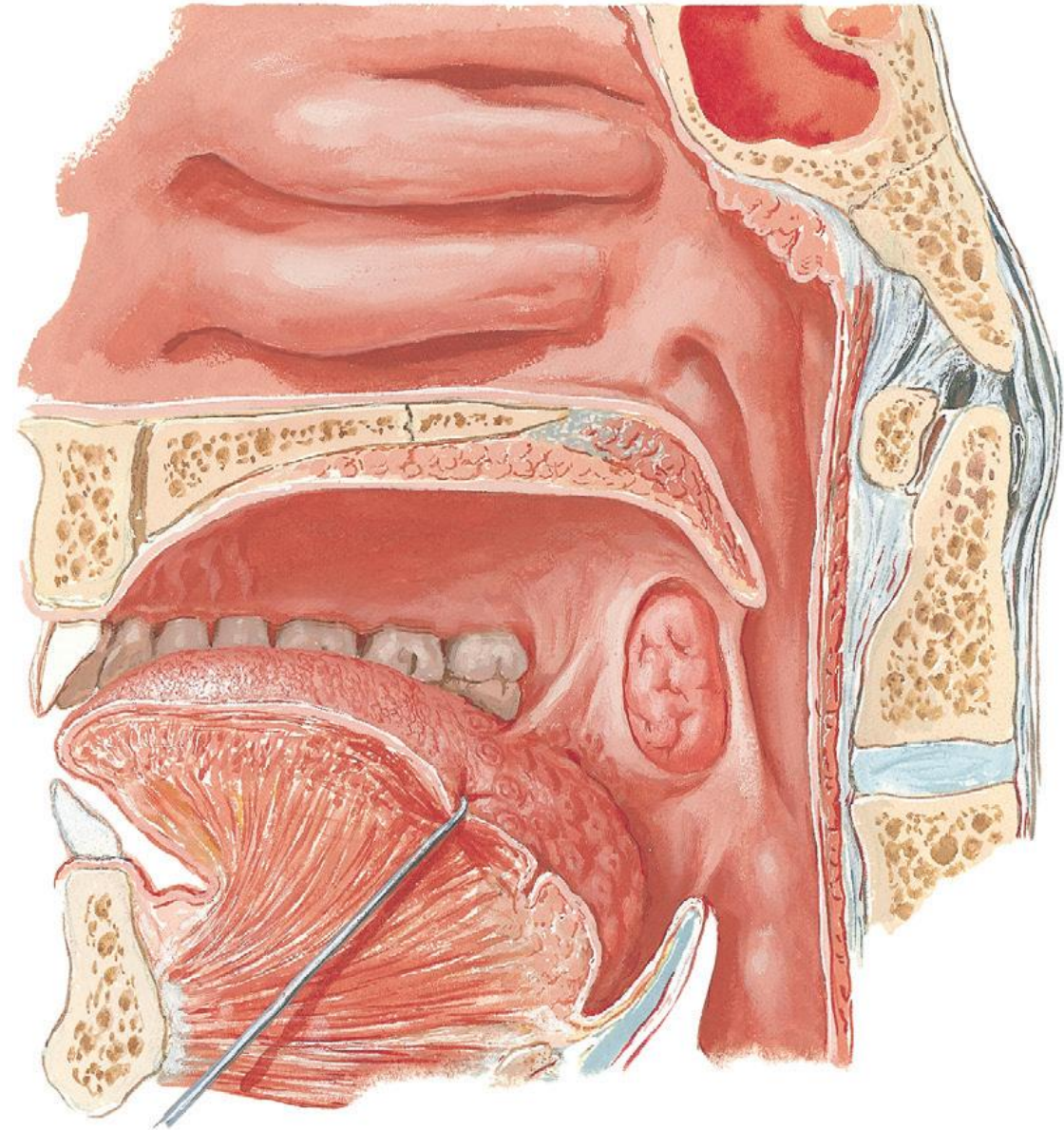
Oral Cavity: Definition & Subdivisions

1. The oral cavity is the first part of GIT
 2. It extends from the oral fissure anteriorly to the oropharynx posteriorly.
- ❑ Two anatomical subdivisions:
- A. Oral vestibule – space between lips/cheeks and teeth/gingiva.
 - B. Oral cavity proper – space within the dental arches.
- Posteriorly, the oral cavity communicates with the oropharynx through the oropharyngeal isthmus (palatoglossal fold=P).



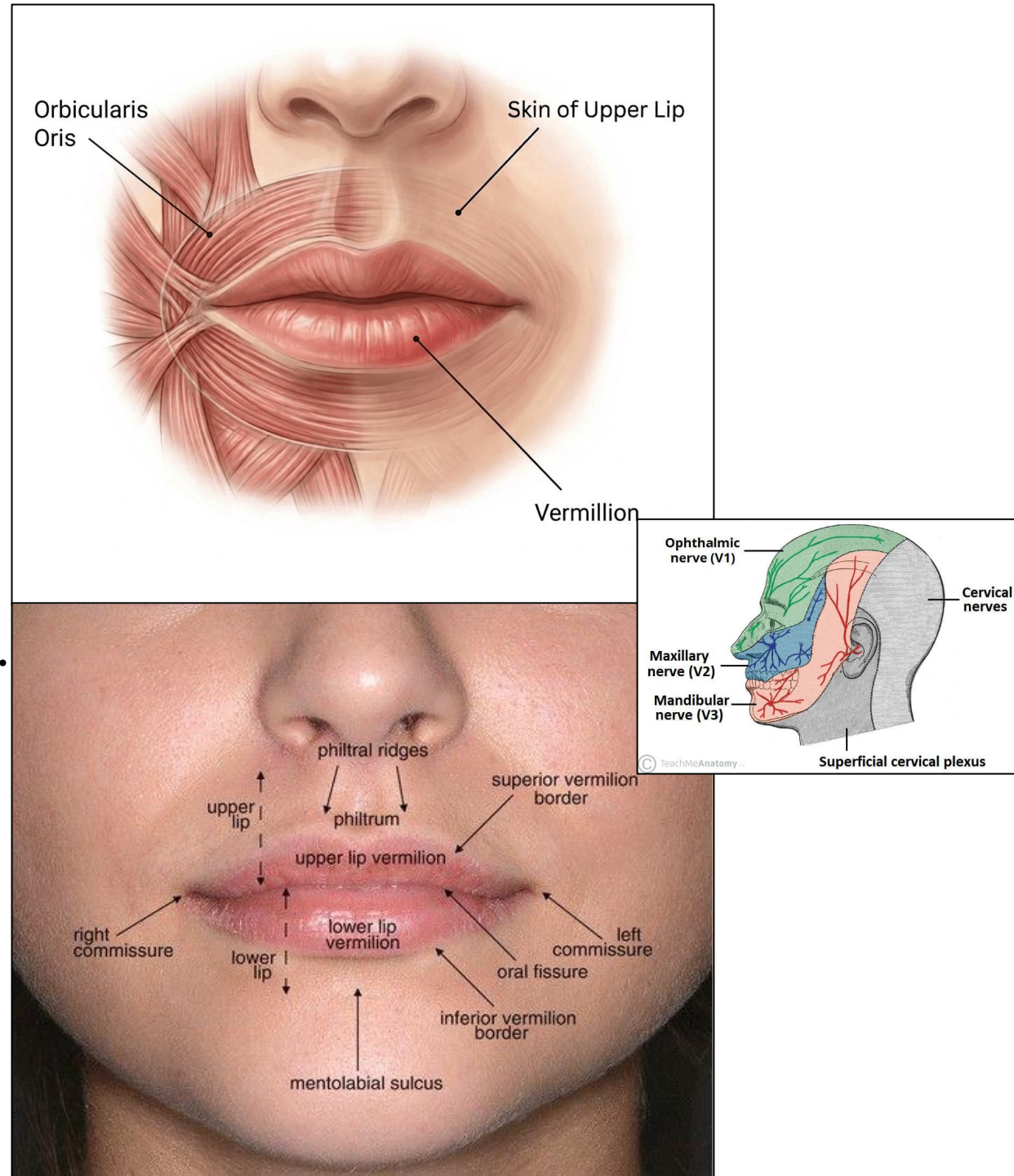
Boundaries of the Oral Cavity

1. Anterior boundary: lips.
 2. Lateral boundaries: cheeks.
 3. Roof: hard palate and soft palate.
 4. Floor: tongue and mucosa over the mylohyoid region.
 5. Posteriorly: opens into the oropharynx (palatoglossal fold).
- Teeth and gingiva form the medial boundary of the vestibule.



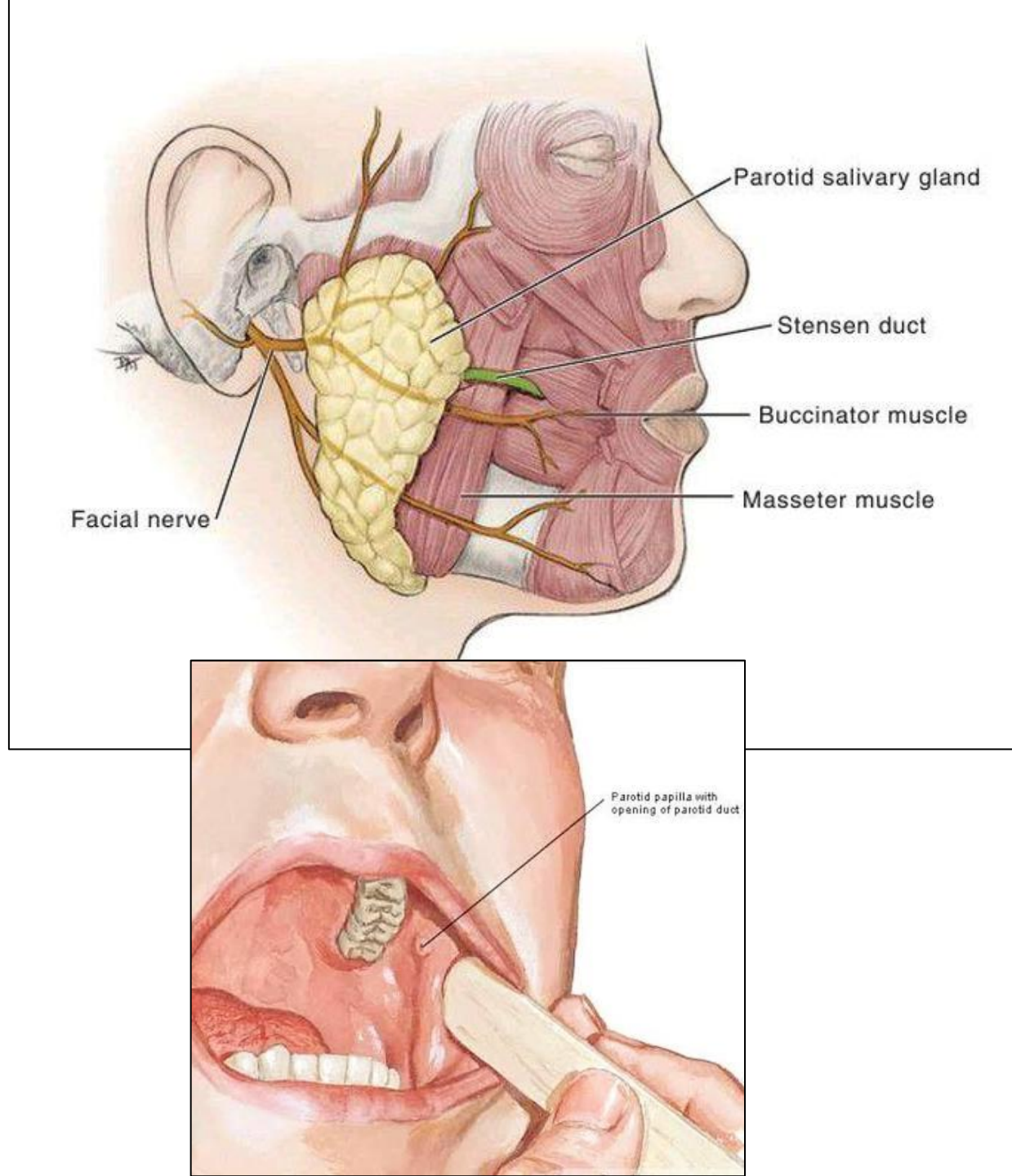
Lips

- Muscular folds forming the anterior boundary of the mouth.
 - Main muscle: **Orbicularis Oris**.
 - Layers: skin → subcutaneous tissue → muscle → glands → oral mucosa.
1. Motor supply: **Facial Nerve (CN VII)**.
 2. Upper lip sensory supply:
Infraorbital Nerve (V2).
 3. Lower lip sensory supply:
Mental Nerve (V3).



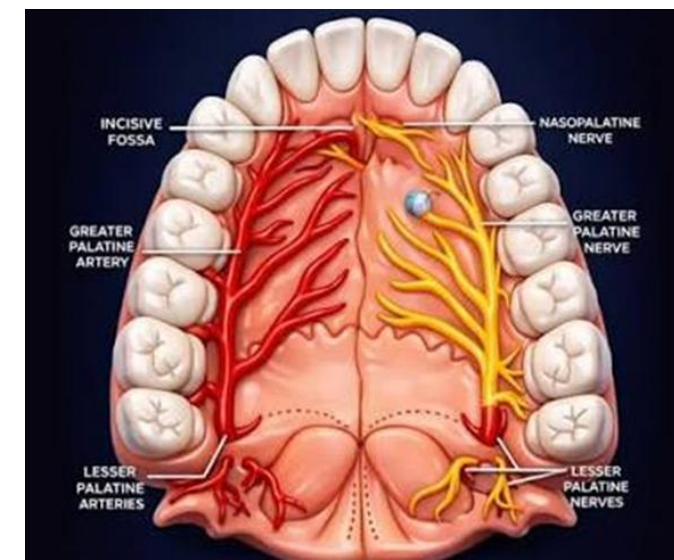
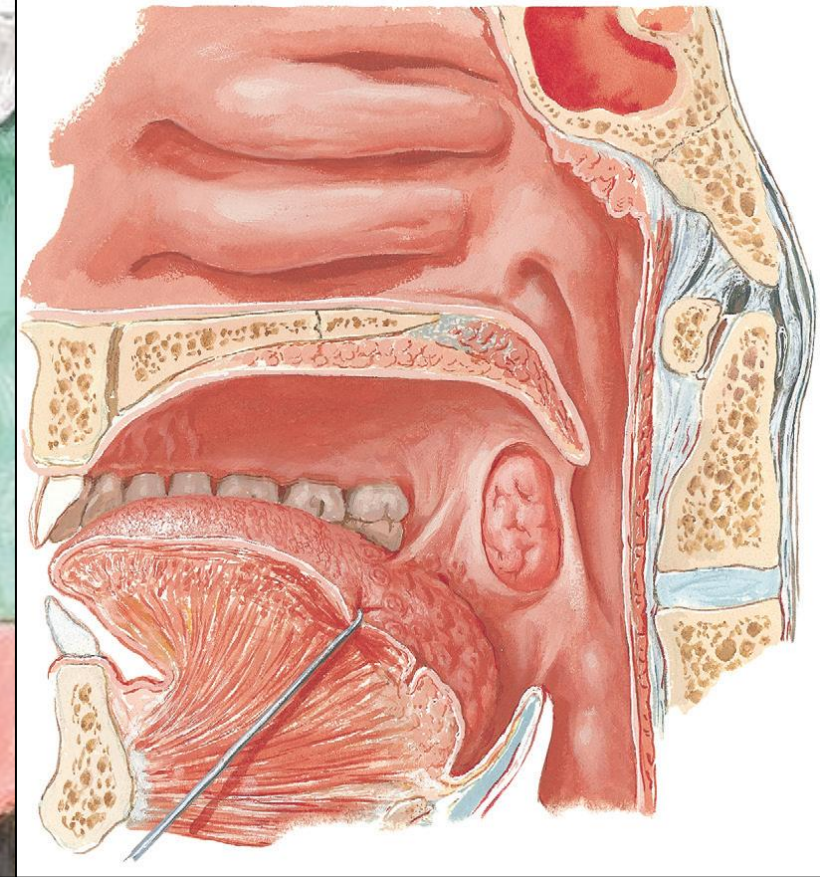
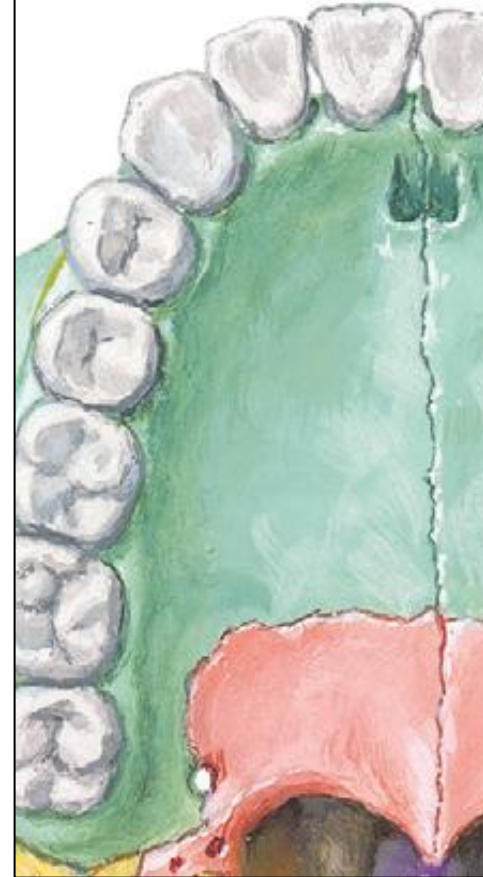
Cheeks & Buccinator

1. The cheek forms the lateral wall of the oral cavity.
2. Main muscle: **Buccinator**.
 - Functions: compresses the cheek against the teeth and assists mastication, sucking and blowing.
 - The parotid duct crosses the masseter, turns medially, pierces buccinator, and opens into the vestibule.
 - Parotid duct opening: opposite the maxillary (upper) second molar.



Palate: Classification

- The palate forms the roof of the oral cavity.
 1. Hard palate – anterior two-thirds; bony and relatively immobile.
 2. Soft palate – posterior one-third; muscular and mobile.
- Hard palate is formed mainly by **palatine processes of maxillae** and **horizontal plates of palatine bones**.
- Soft palate closes the nasopharynx during swallowing.

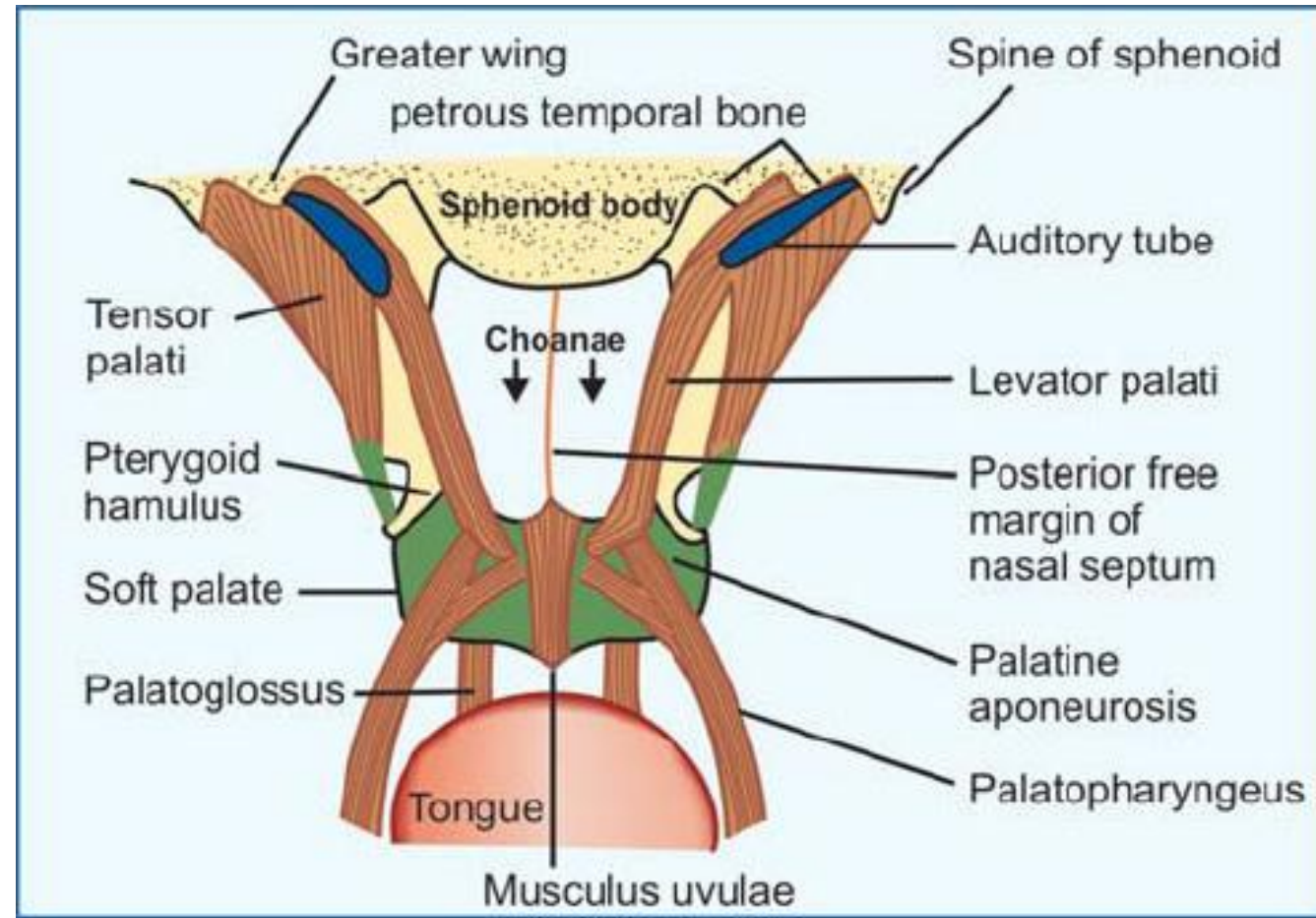


Soft Palate Muscles:

1. **Tensor veli palatini** (Palatine Aponeurosis is formed mainly muscle)
2. **Levator veli palatini**
3. **Palatoglossus**
4. **Palatopharyngeus**
5. **Musculus uvulae**

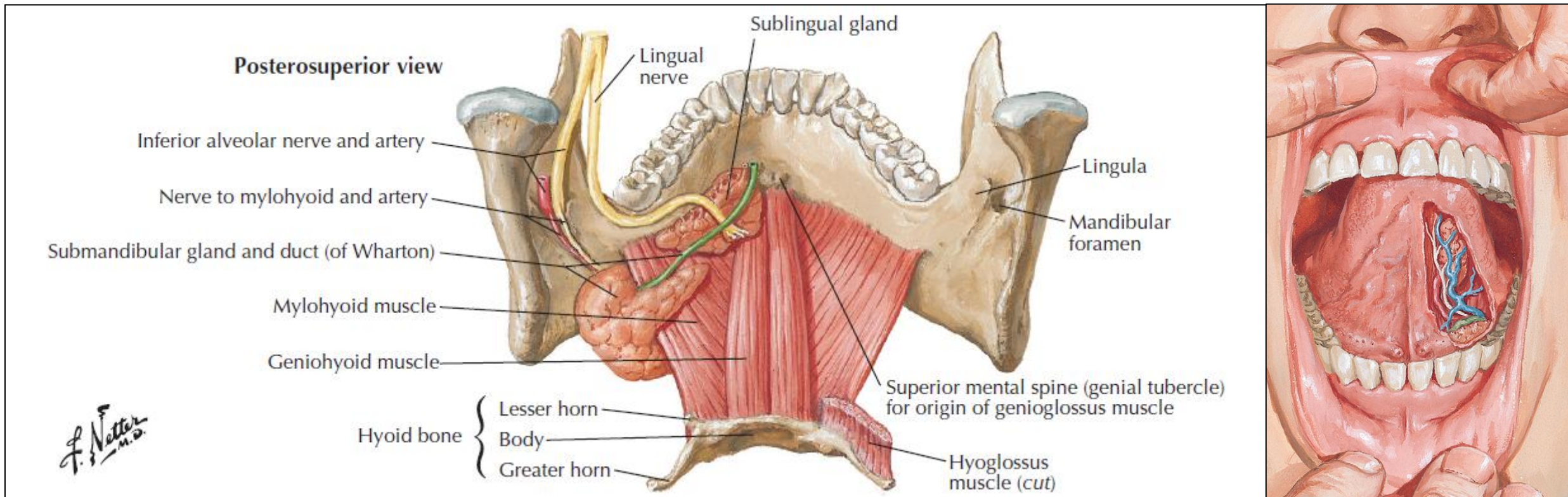
❑ Motor Innervation:

- Most muscles by **Vagus Nerve (CN X)** through the pharyngeal plexus.
- Exception: tensor veli palatini → **Mandibular Nerve (V3).**



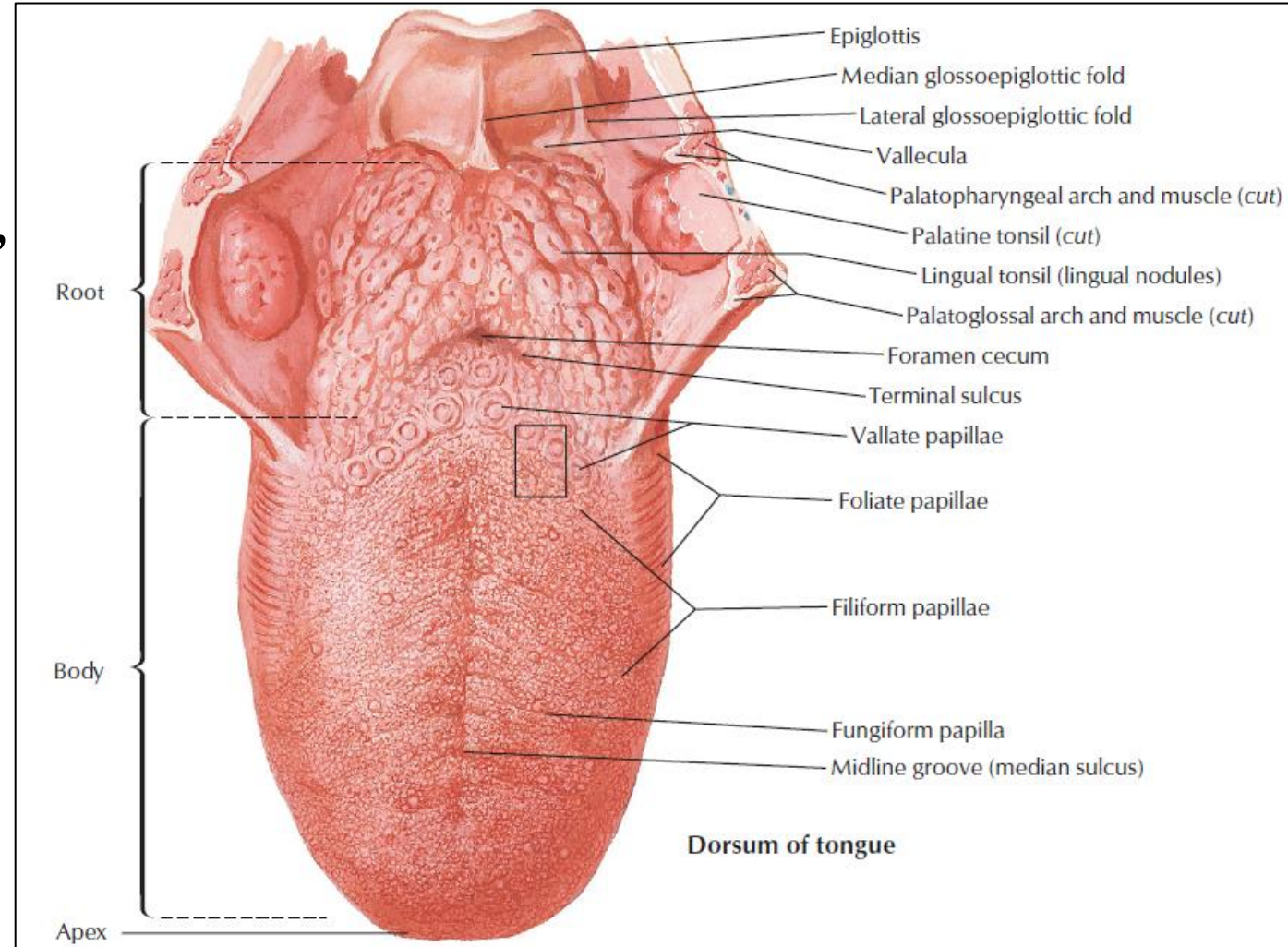
Floor of the Mouth

1. Main muscular support: **Mylohyoid Muscles.**
2. Important structures: tongue, sublingual glands, and submandibular ducts.
3. Submandibular duct opens on the sublingual caruncle beside the lingual frenulum.
4. Sublingual gland has numerous ducts opening along the sublingual fold.



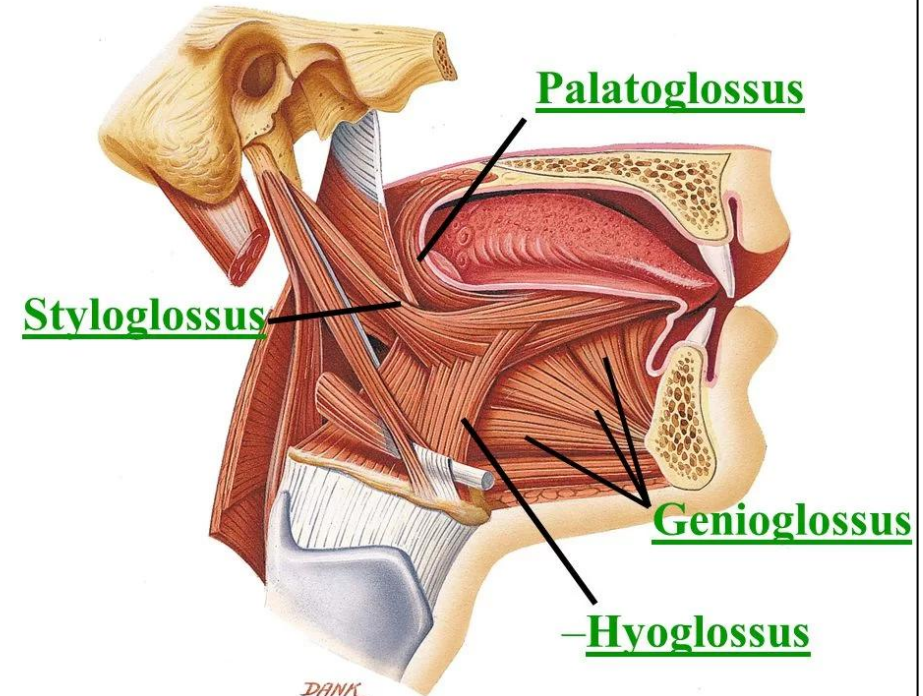
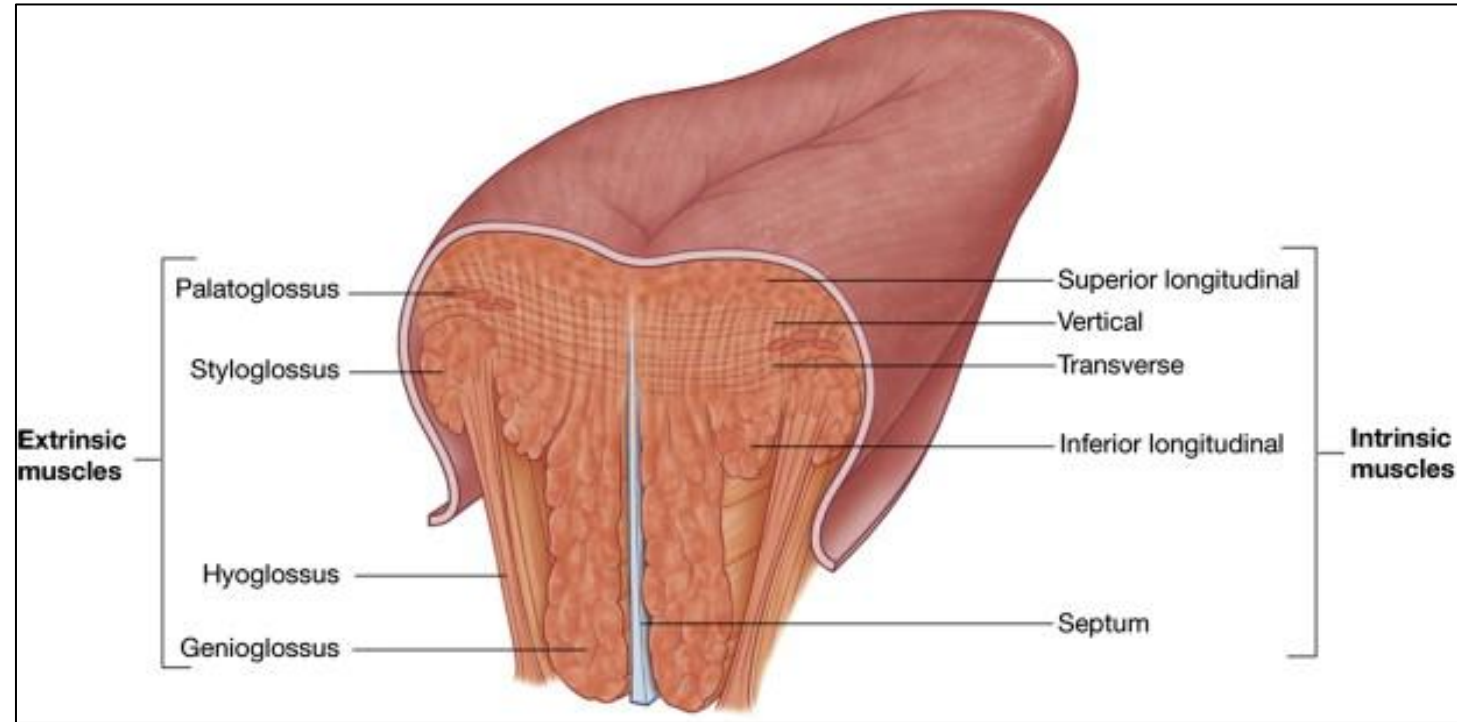
Tongue:

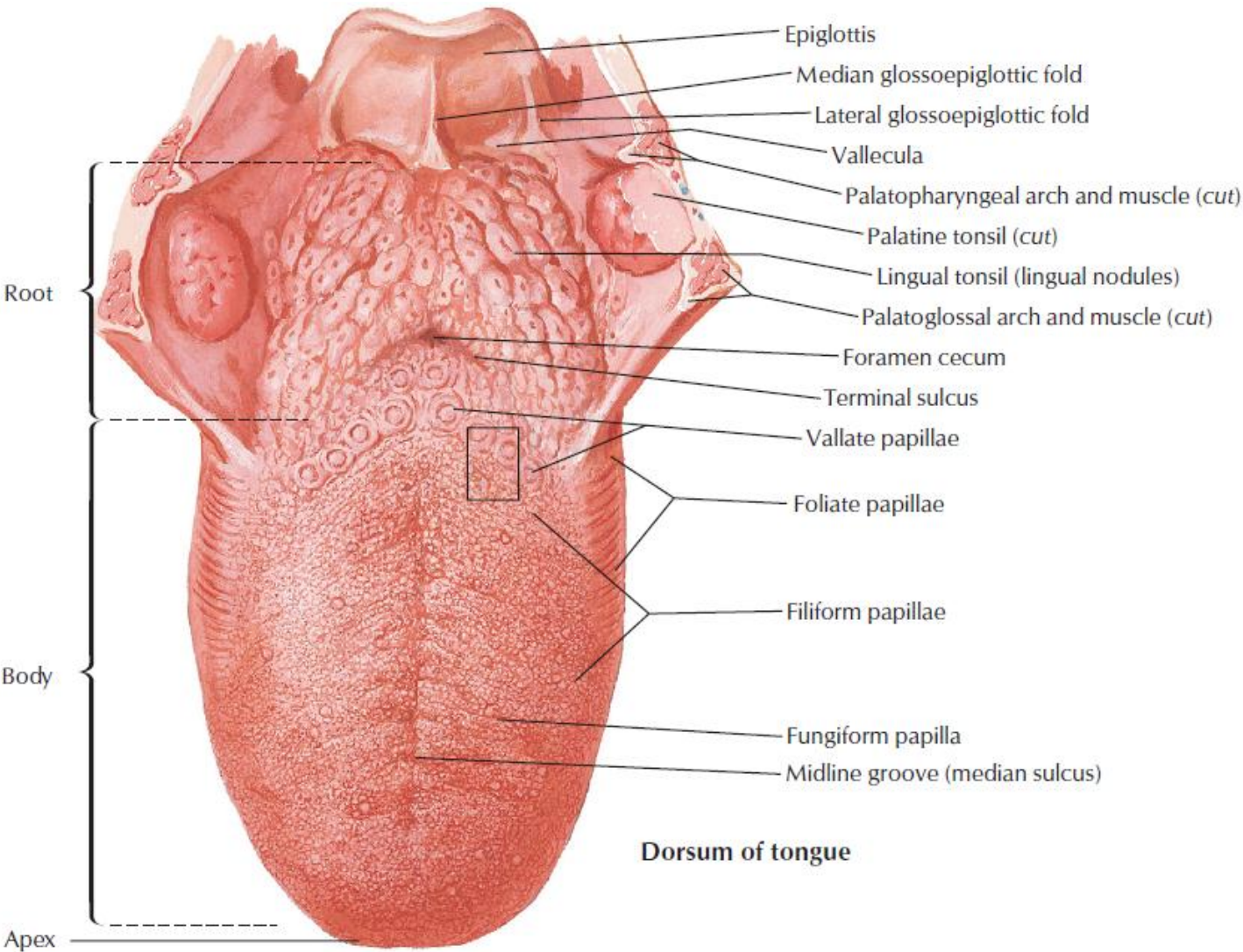
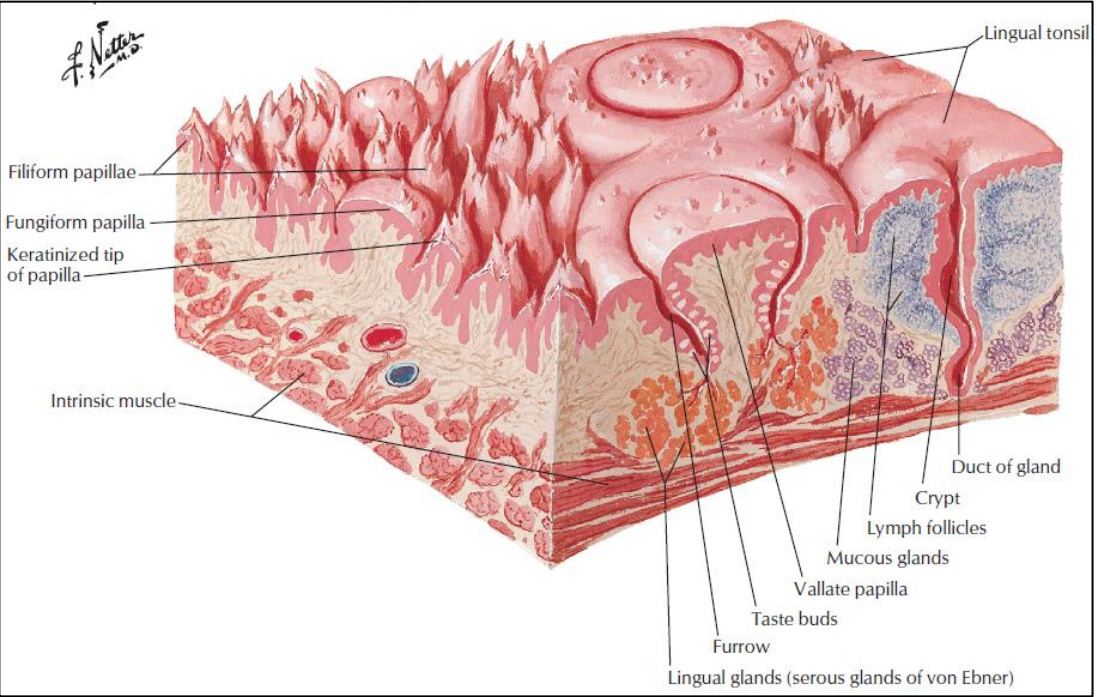
1. The tongue is a muscular organ involved in taste, speech, mastication, and swallowing.
2. Anterior 2/3: oral part.
3. Posterior 1/3: oropharyngeal part.
4. The two parts are separated by the **Sulcus Terminalis**.
5. **Foramen Cecum** lies at the apex of the represents an embryological remnant of the thyroglossal duct.



Tongue Muscles

1. Intrinsic muscles: superior longitudinal, inferior longitudinal, transverse, and vertical.
2. Extrinsic muscles: genioglossus, hyoglossus, styloglossus, and palatoglossus.
3. Most tongue muscles: motor supply by **Hypoglossal Nerve (CN XII)**.
4. Exception: palatoglossus → **Vagus Nerve (CN X)**.
5. Intrinsic muscles change the shape of the tongue; extrinsic muscles move it.



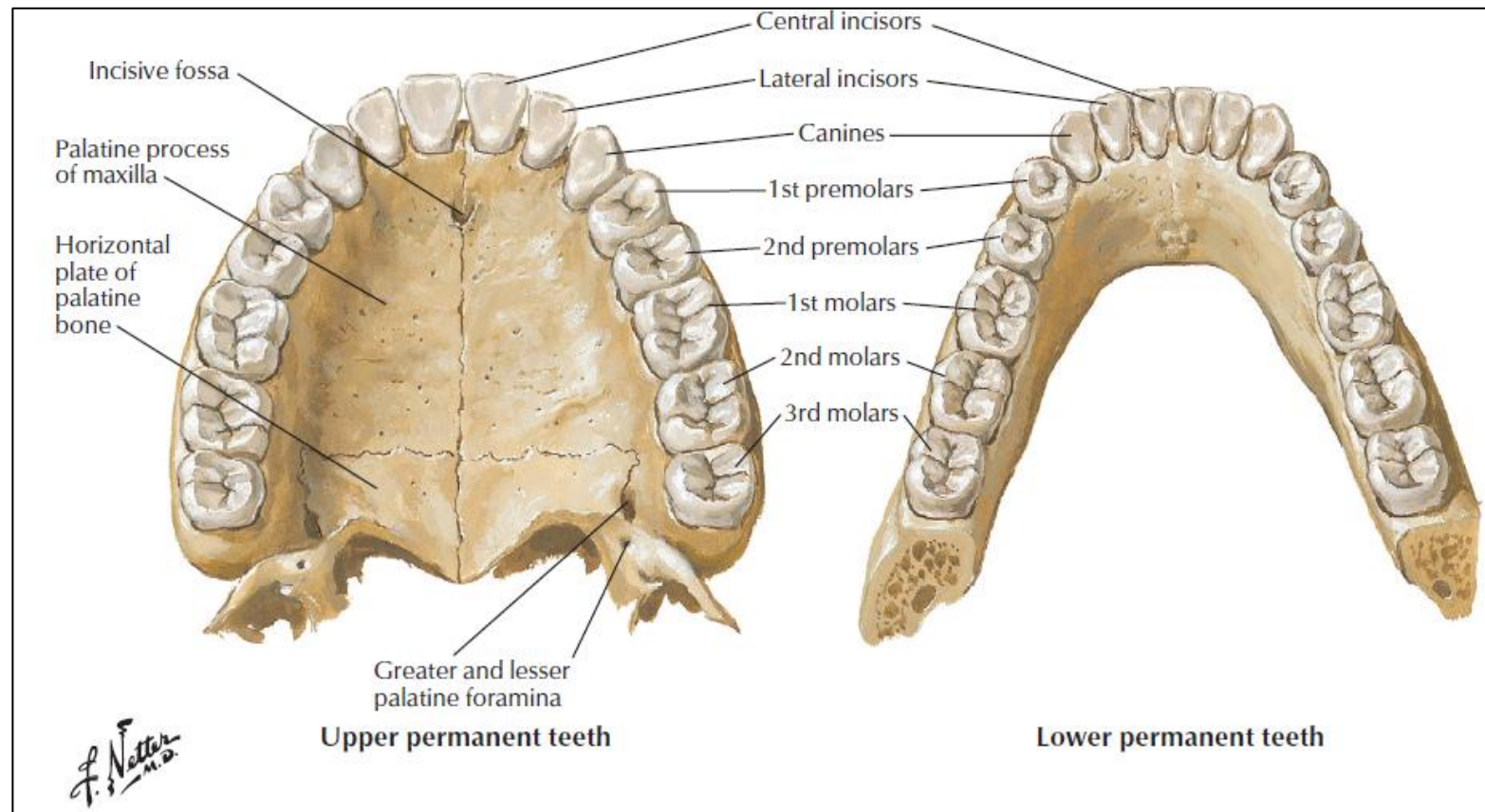


Teeth: Classification

❑ Adult permanent dentition contains 32 teeth.

1. Incisors – cutting.
2. Canines – tearing.
3. Premolars – crushing.
4. Molars – grinding.

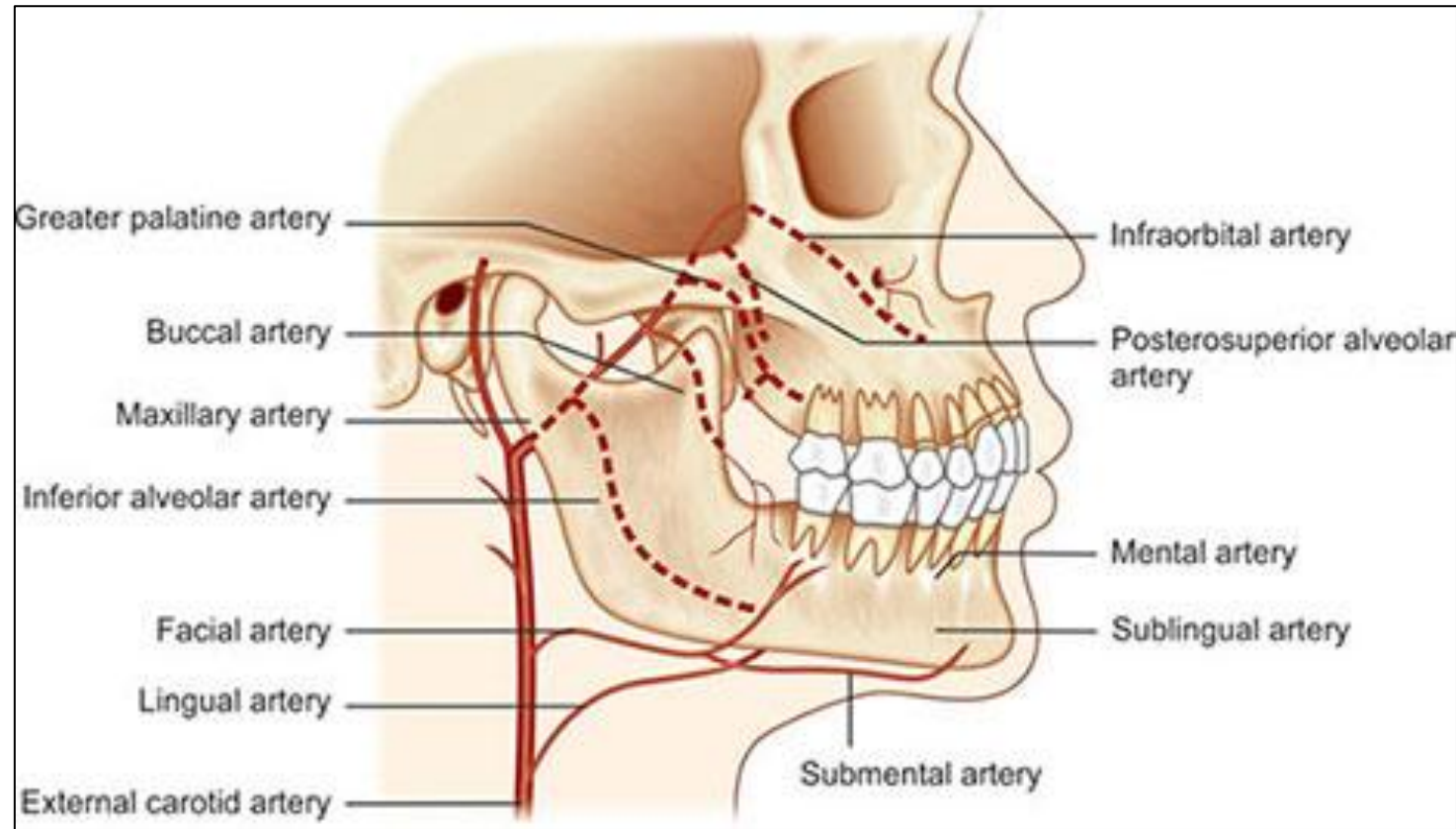
❑ Primary (deciduous) dentition contains 20 teeth (Note: Deciduous dentition does **not** include premolars (bicuspid) or third molars (wisdom teeth).)



Blood Supply

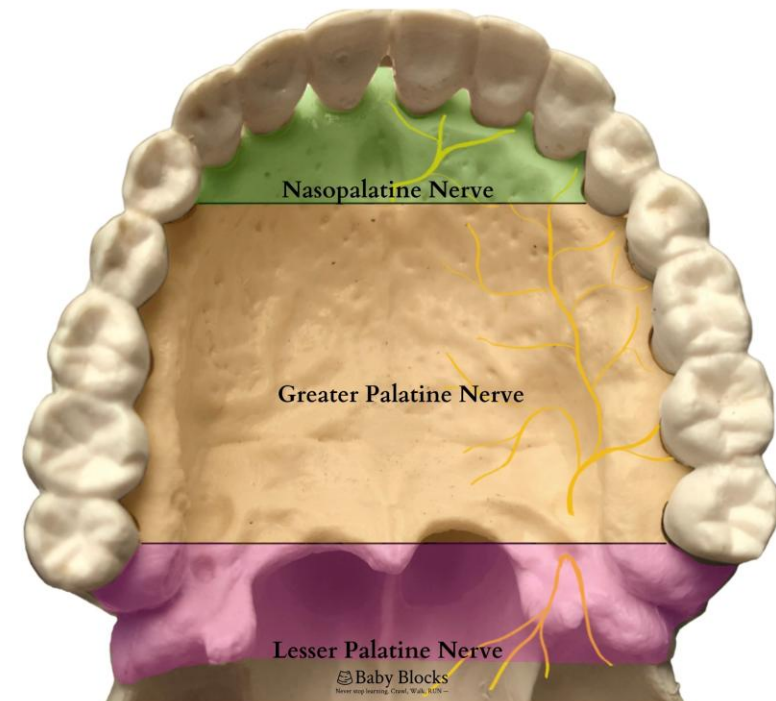
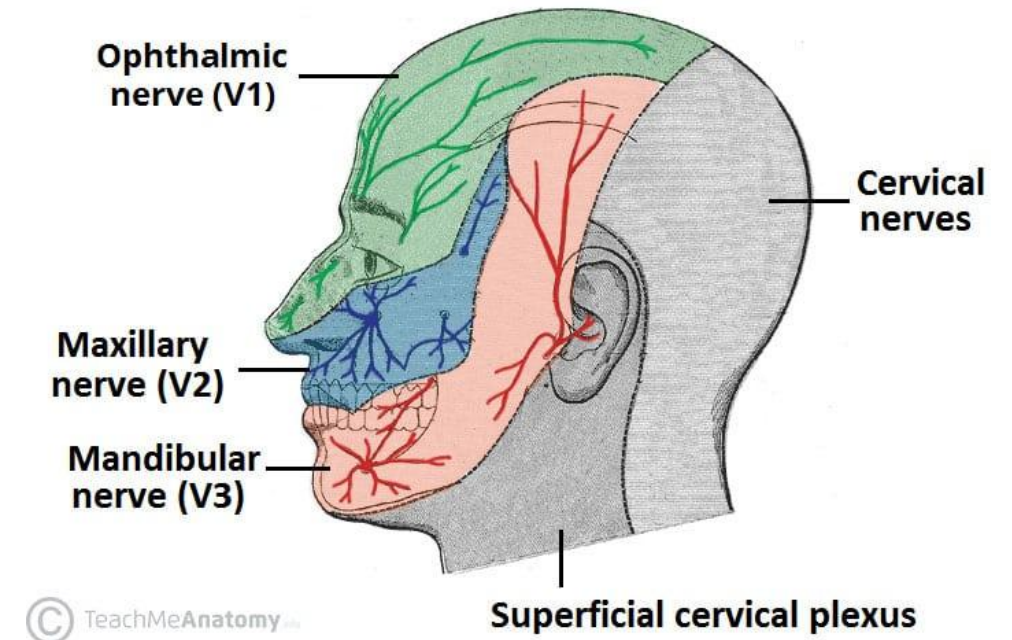
❑ The oral cavity receives blood mainly from branches of the external carotid artery.

1. **Lingual artery** → tongue and floor of mouth.
 2. **Facial artery** → lips and parts of the cheek/palate.
 3. **Maxillary artery** → teeth, palate, and deeper oral structures.
- ❖ Rich vascularity is clinically important for healing and bleeding.

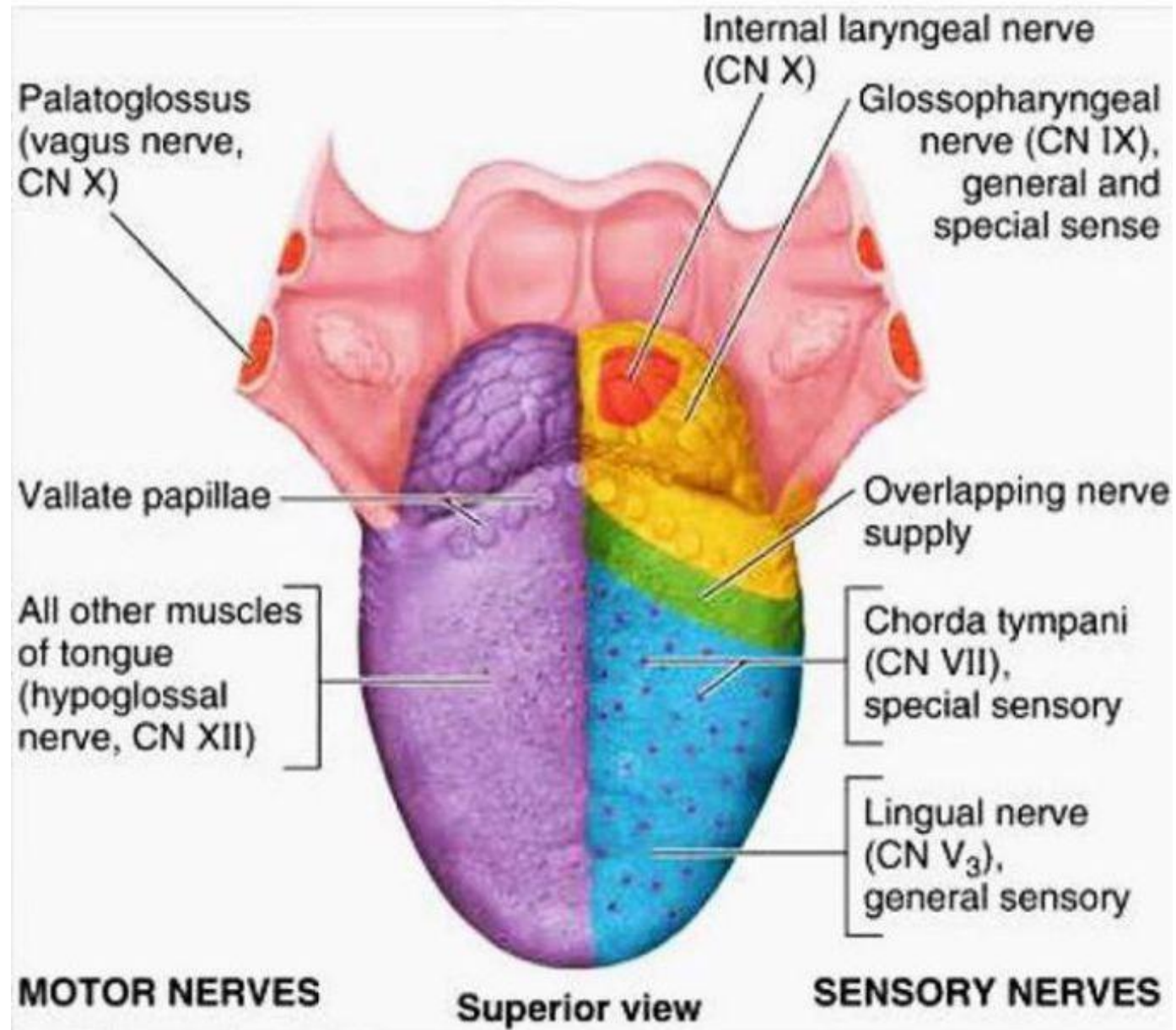


Sensory Innervation

1. Most general sensory supply is derived from the **Trigeminal Nerve (CN V)**.
 2. Upper lip → infraorbital nerve (V2).
 3. Lower lip → mental nerve (V3).
 4. Buccal mucosa → buccal nerve (V3).
- Hard palate → greater palatine, lesser palatine and nasopalatine nerves.

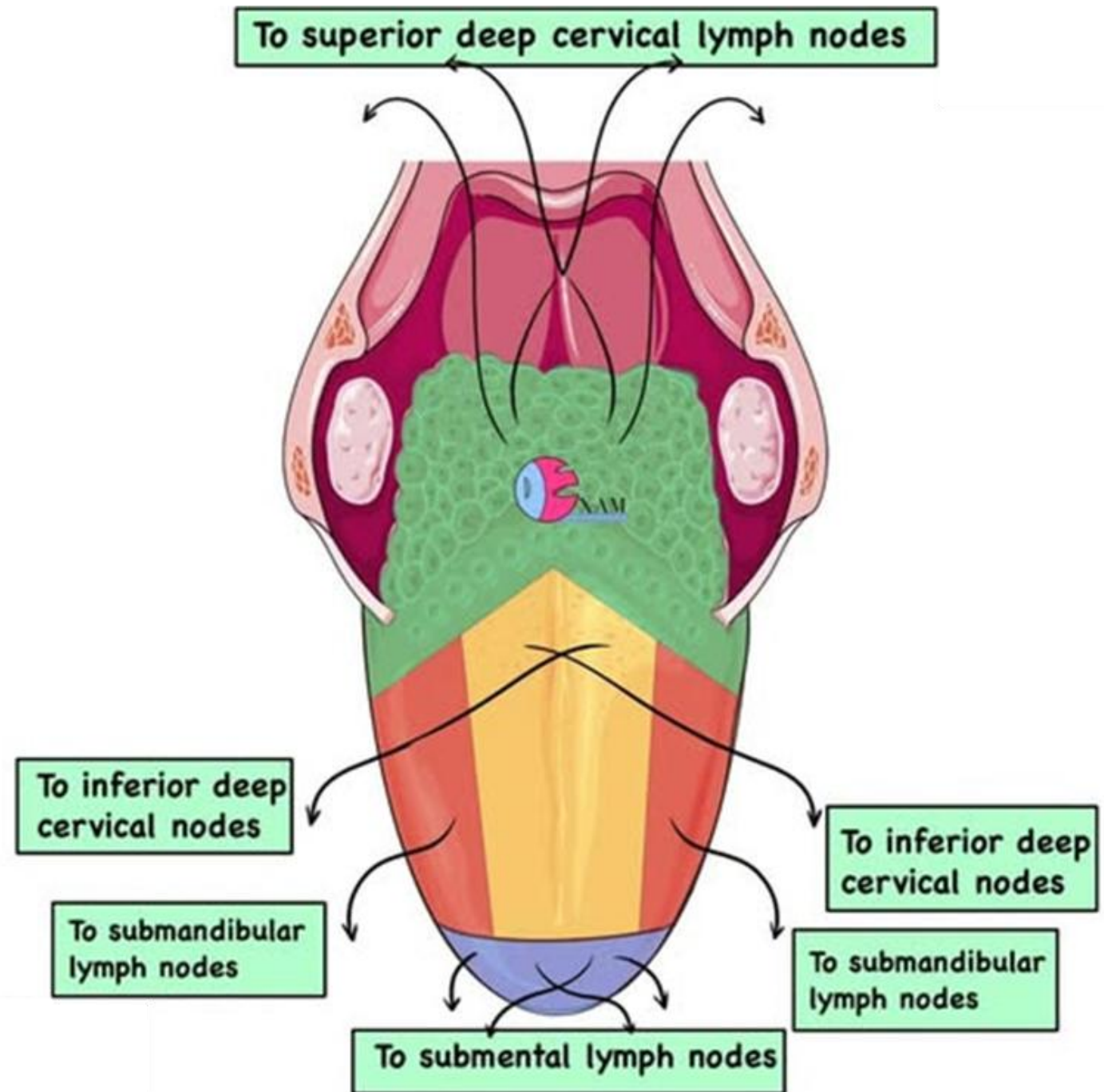


Innervation of the Tongue



Lymphatic Drainage of Tongue

- Major groups: submental, submandibular, and upper deep cervical lymph nodes.
- Tip of tongue → mainly submental nodes.
- Lateral anterior tongue → mainly submandibular nodes.
- Central tongue has important bilateral drainage to deep cervical nodes.
- Clinical importance: oral cancers may metastasize early to cervical lymph nodes.



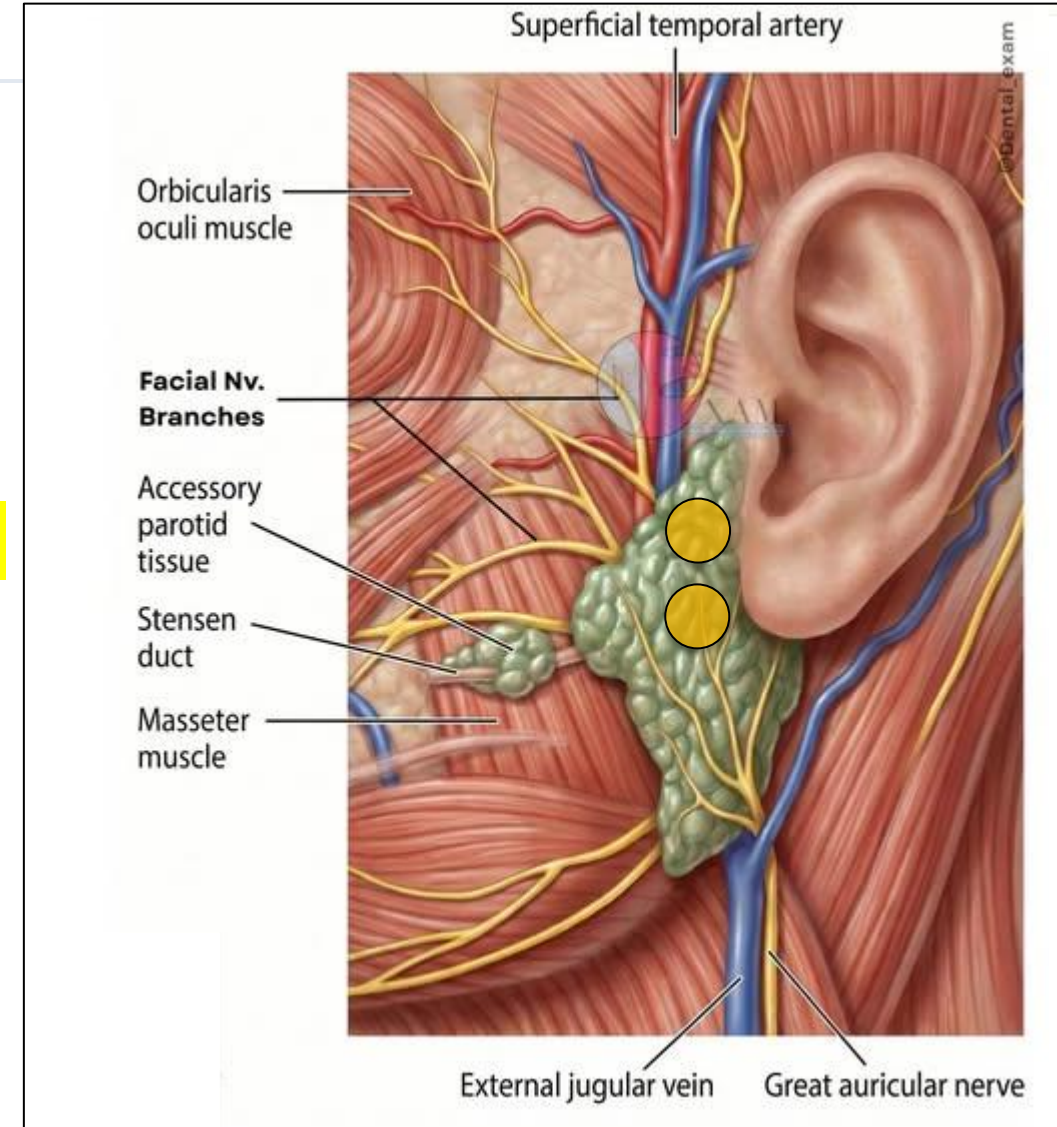
Anatomy of Salivary Glands

Salivary Glands :

- **Functional Classification:** Exocrine glands producing and secreting saliva to facilitate mastication, swallowing, speech, and initial carbohydrate digestion via salivary amylase.
- **Categorization:** Broadly divided into major and minor salivary glands.
 1. **Major Salivary Glands:** Three paired macroscopic structures (parotid, submandibular, and sublingual) responsible for roughly 90% to 95% of total daily salivary volume.
 2. **Minor Salivary Glands:** Hundreds of microscopic glands scattered throughout the submucosa of the lips, cheeks, palate, and tongue.

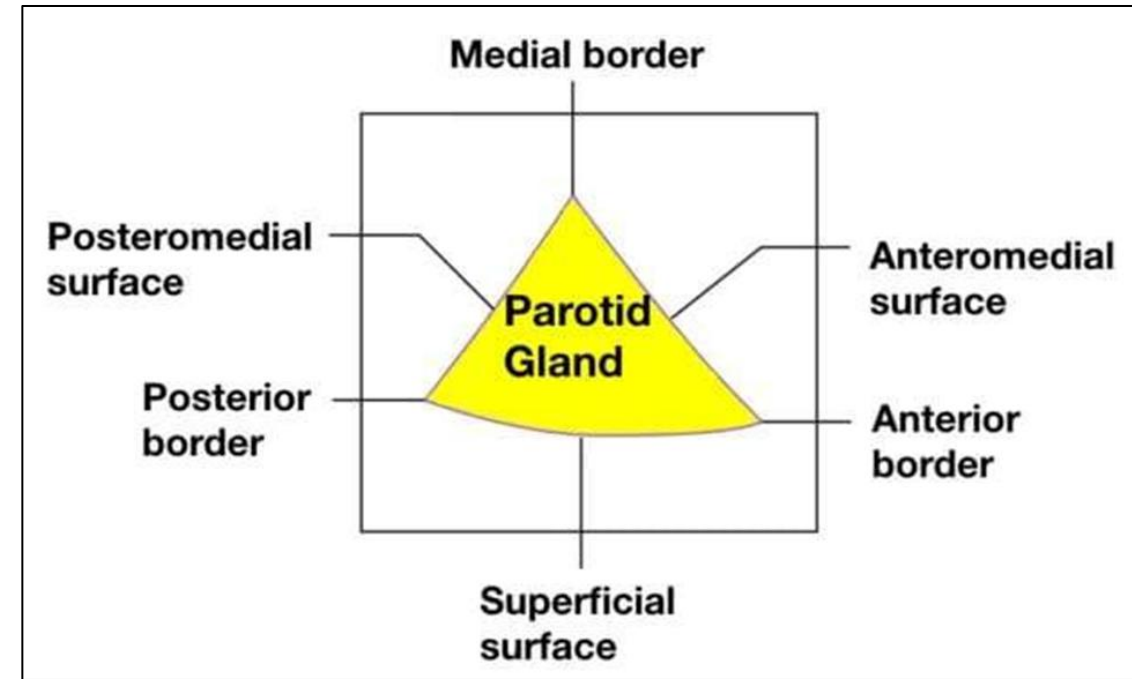
1. Parotid Gland:

- The largest major salivary gland. A serous-type gland enclosed within a tough layer of deep cervical fascia known as the parotid sheath.
- Location and Shape: Occupies the **Retromandibular Fossa**, wedged between the mandibular ramus anteriorly and the mastoid process and sternocleidomastoid muscle posteriorly.



The Parotid Gland Borders:

- **Anterior Border:** Separates the superficial surface from the anteromedial surface, resting against the posterior border of the mandibular ramus. The **Branches of Facial Nerve** and the **Parotid (Stensen's) Duct** emerge from or cross over this border.

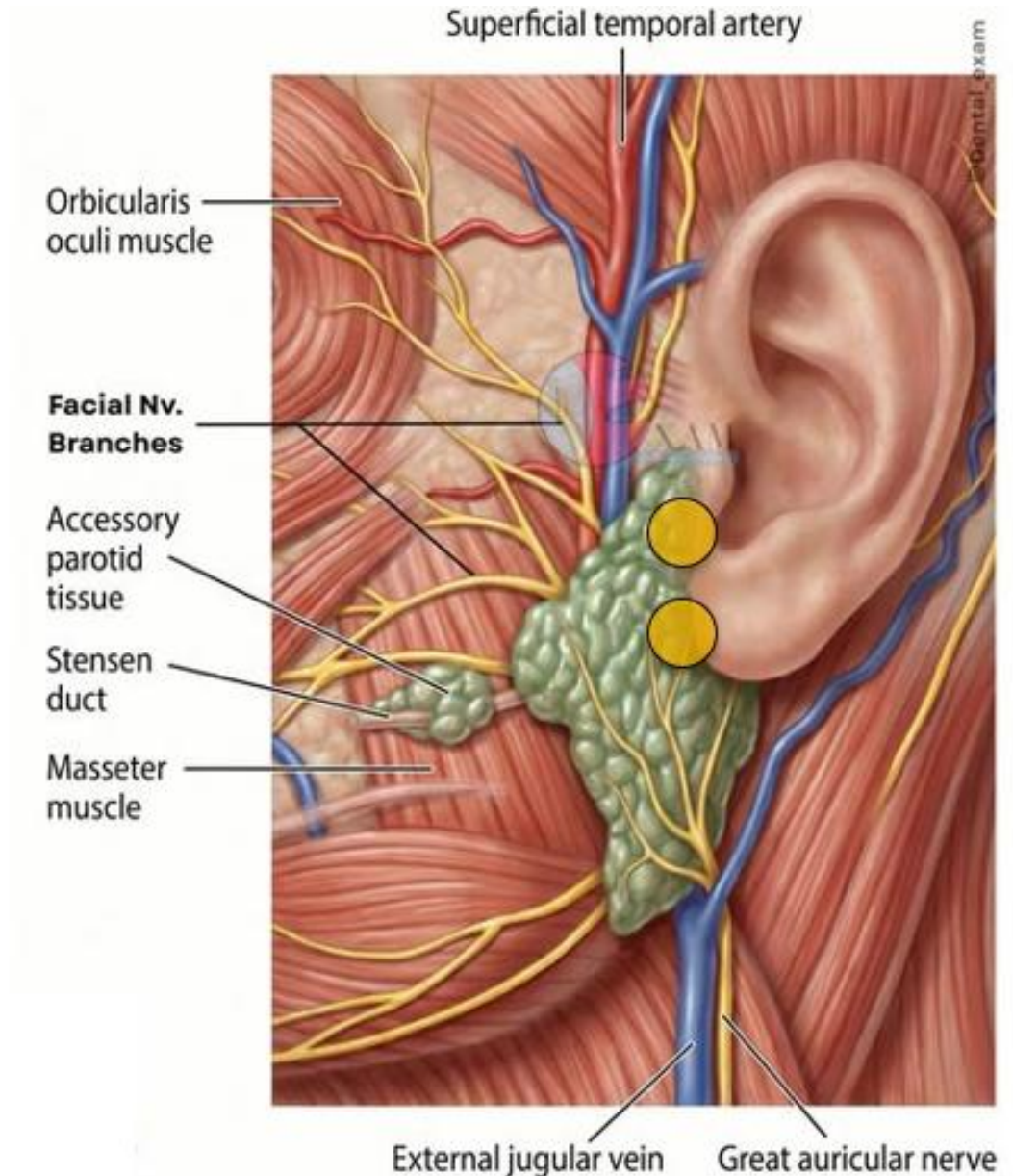


- **Posterior Border:** Separates the superficial surface from the posteromedial surface, overlapping and resting against the anterior border of the **Sternocleidomastoid Muscle**.
- **Medial Border:** Formed where the anteromedial and posteromedial surfaces converge deeply, pointing toward the **Lateral Pharyngeal Wall** and wrapping around the styloid process and **Carotid Sheath** structures.

Surfaces and Relations

1. Superficial (Lateral) Surface:

- **Position:** Faces laterally toward the skin and subcutaneous tissue.
- **Relations:**
 - Covered by the skin, superficial fascia, and **platysma muscle**.
 - Contains the **superficial parotid (pre-auricular) lymph nodes** lying directly on its capsule.
 - Crossed by branches of the **great auricular nerve** (supplying the skin over the angle of the mandible and parotid region).

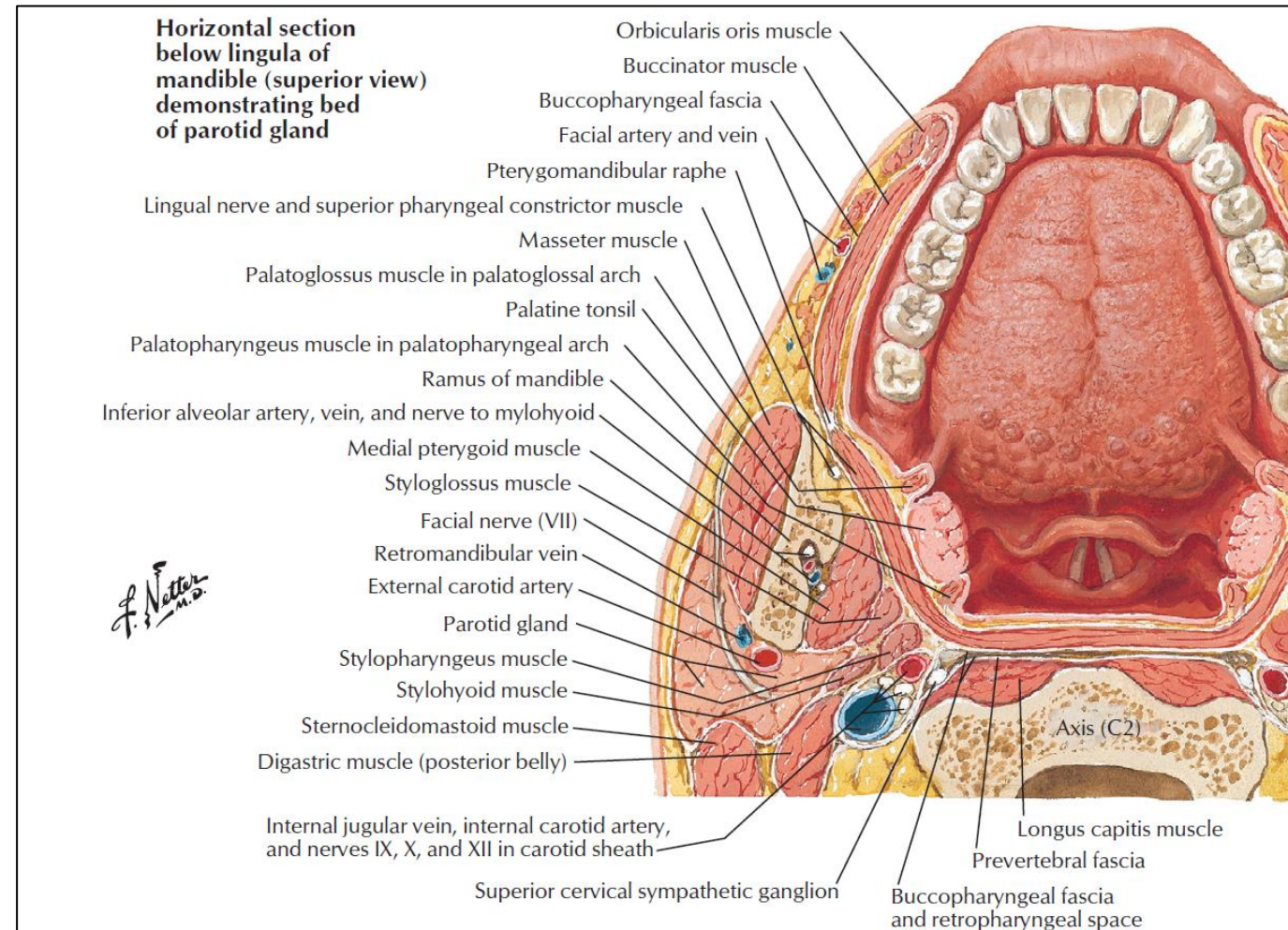
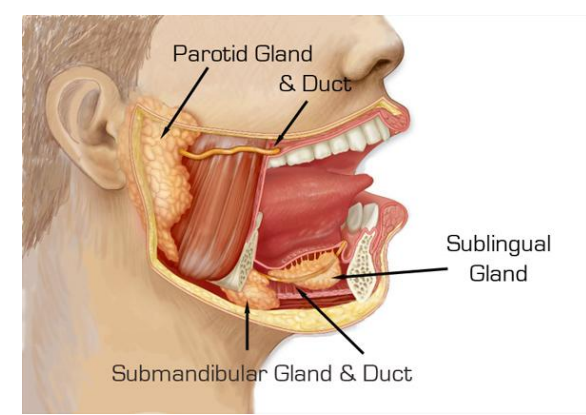


2. Anteromedial Surface

- **Position:** Molded against the posterior border of the mandibular ramus.

- **Relations:**

- **Anteriorly:** Rests against the posterior border of the **ramus of the mandible** and the insertions of the **masseter** and **medial pterygoid** muscles.
- **Emerging Structures:** The branches of the **facial nerve (CN VII)** emerge from beneath the anterior border of this surface to supply the muscles of facial expression.
- **Duct System:** The **parotid (Stensen's) duct** crosses horizontally across the masseter muscle from the upper part of this surface.

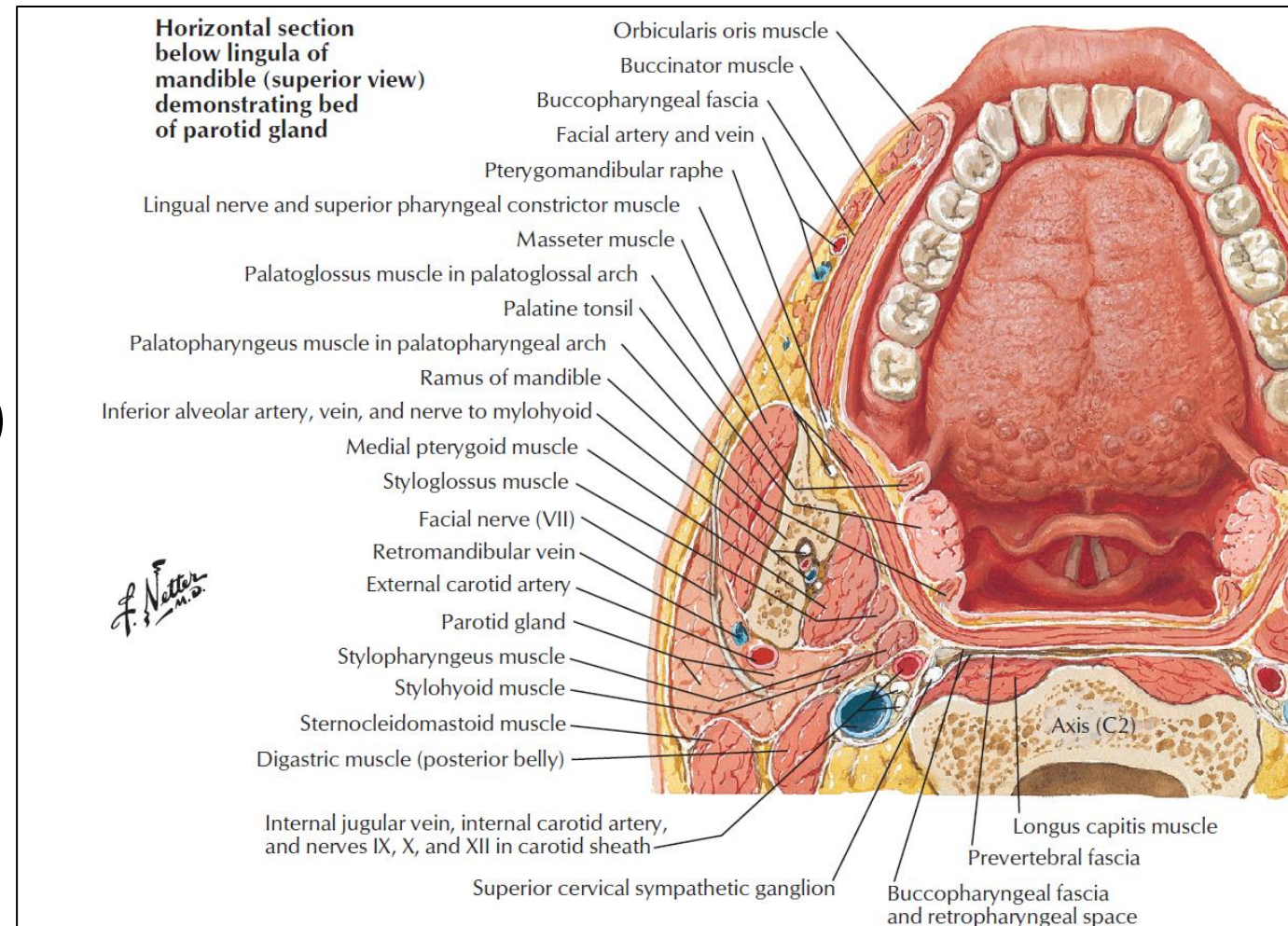


3. Posteromedial Surface

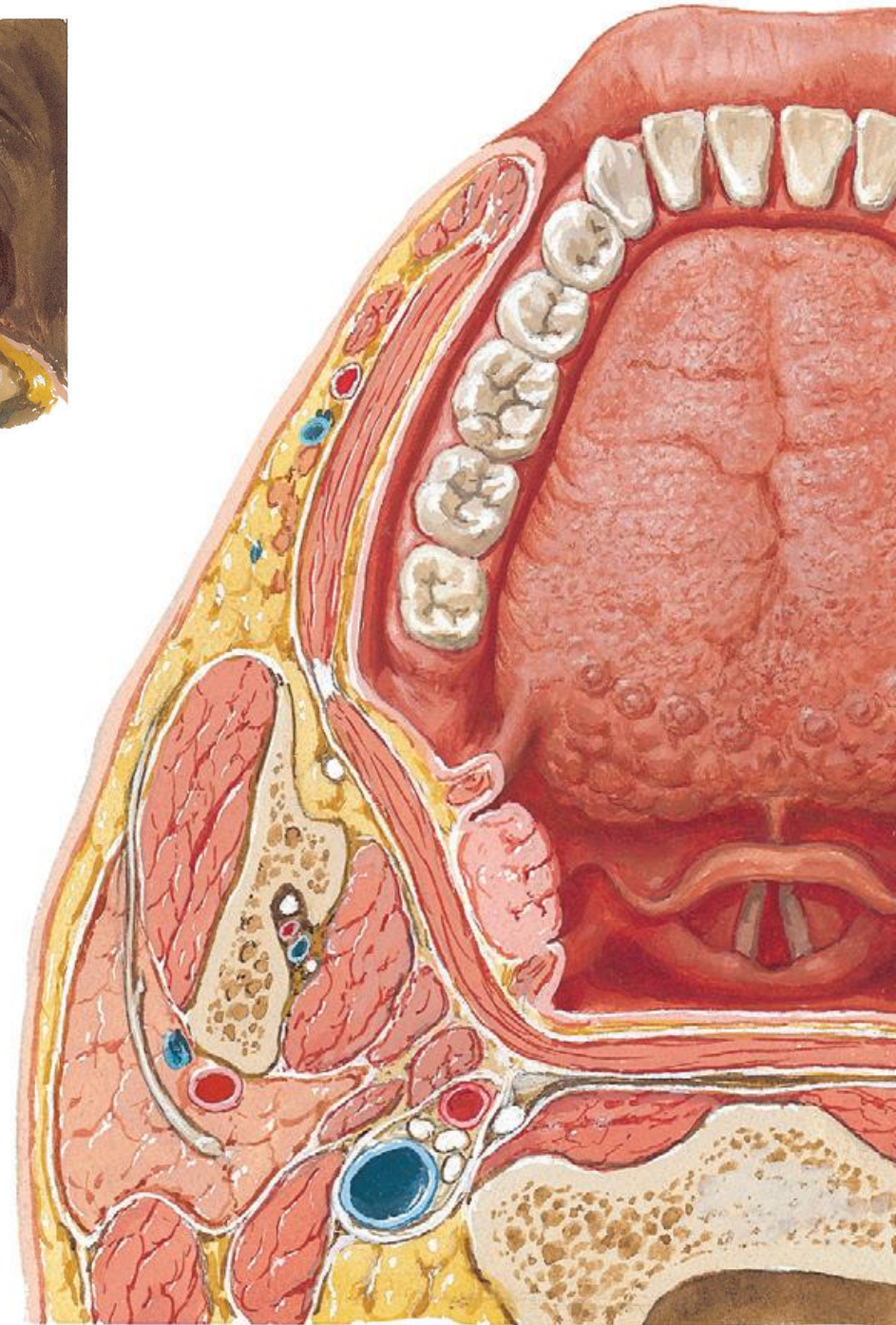
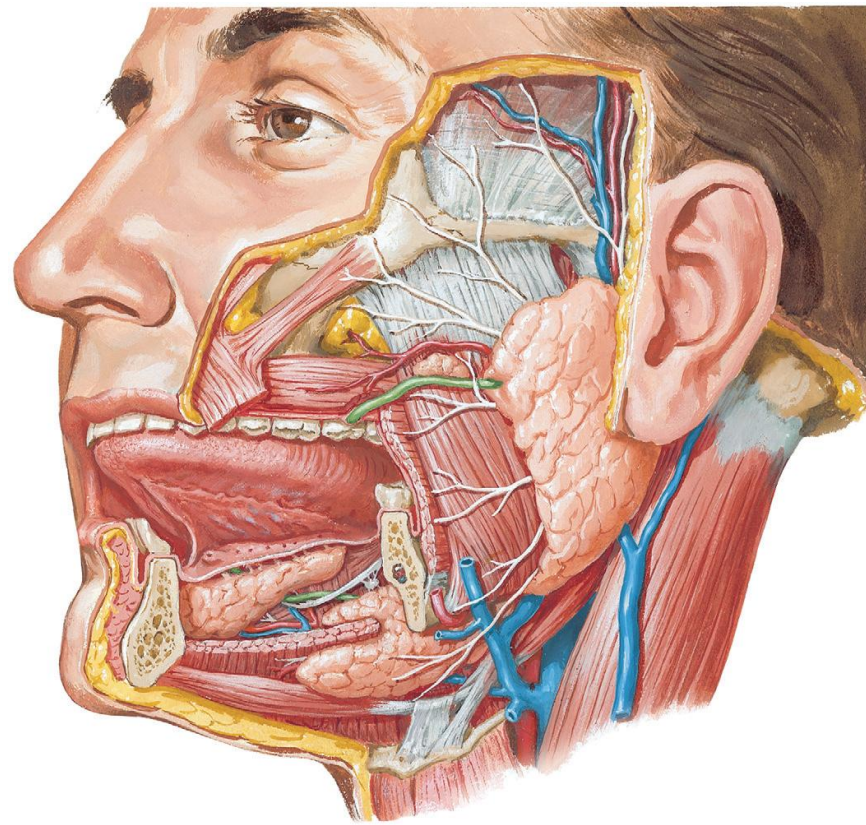
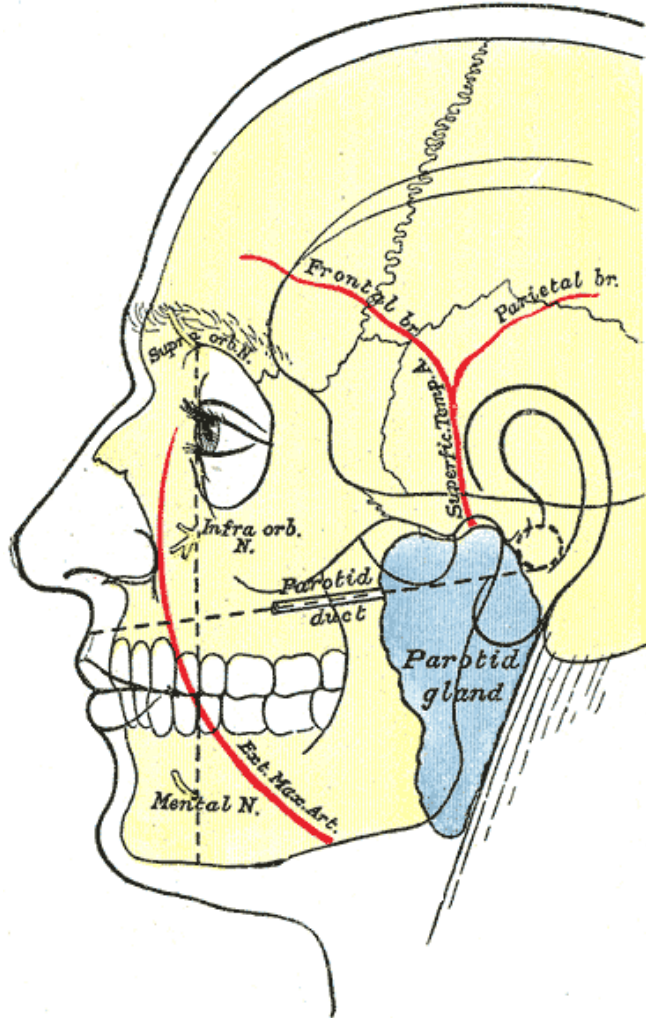
- **Position:** Forms the deep apex of the wedge, facing backward and medially toward the prevertebral region.

- **Relations:**

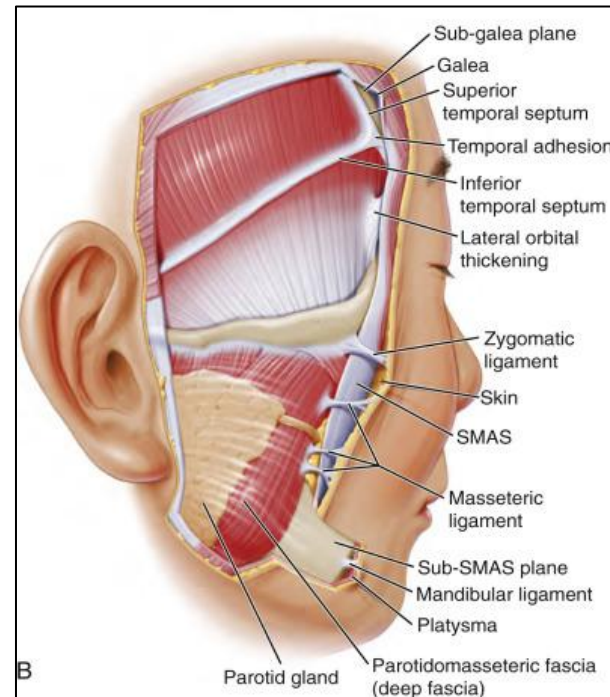
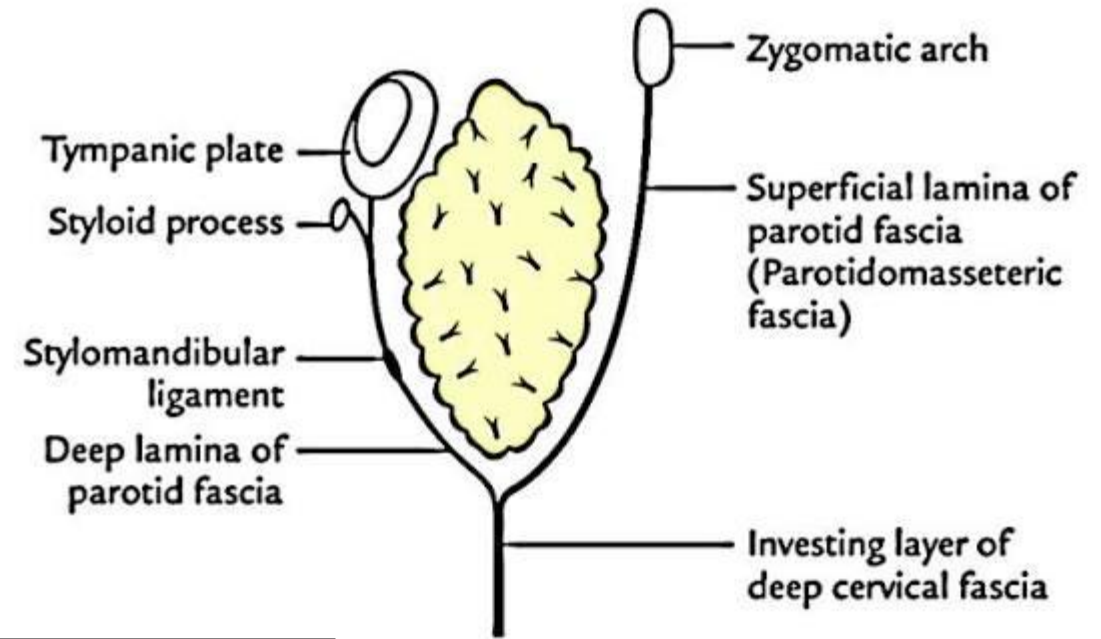
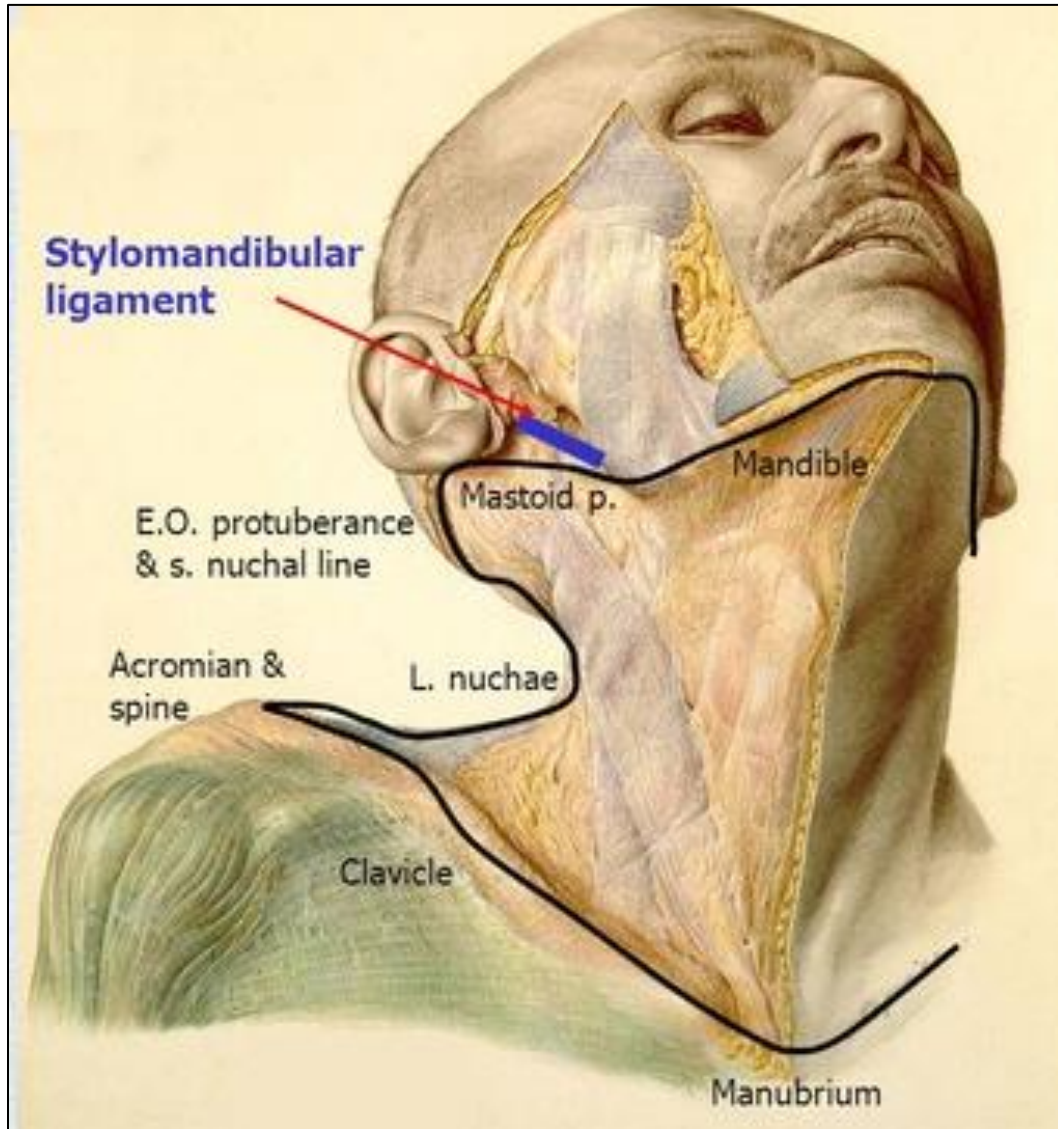
- **Muscles:** Rests against the **sternocleidomastoid** and **posterior belly of the digastric** muscles superiorly, and the **styloid process** with its attached muscles (*styloglossus*, *stylohyoid*, and *stylopharyngeus*).
- **Vessels and Nerves:** Closely related to the **internal carotid artery**, **internal jugular vein**, and the last four cranial nerves (**CN IX, X, XI, and XII**) located deep within the carotid sheath.
- **Tympanic Plate:** Lies against the bony external acoustic meatus superiorly.



Relations



Parotid Capsule



4. Submandibular Gland: Architecture & Surfaces

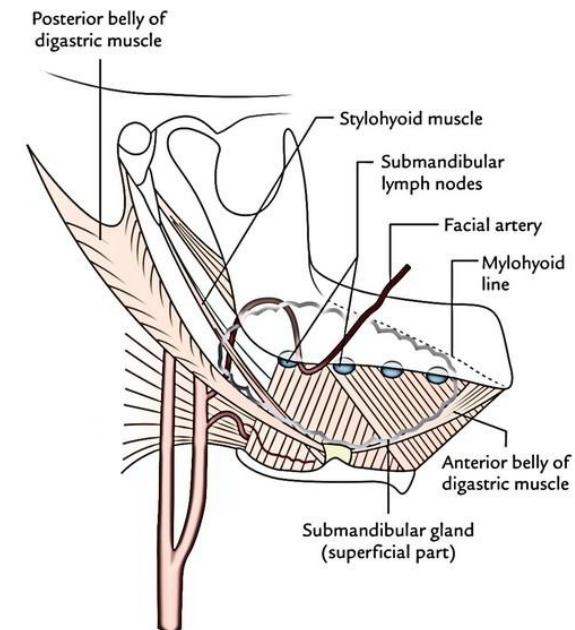
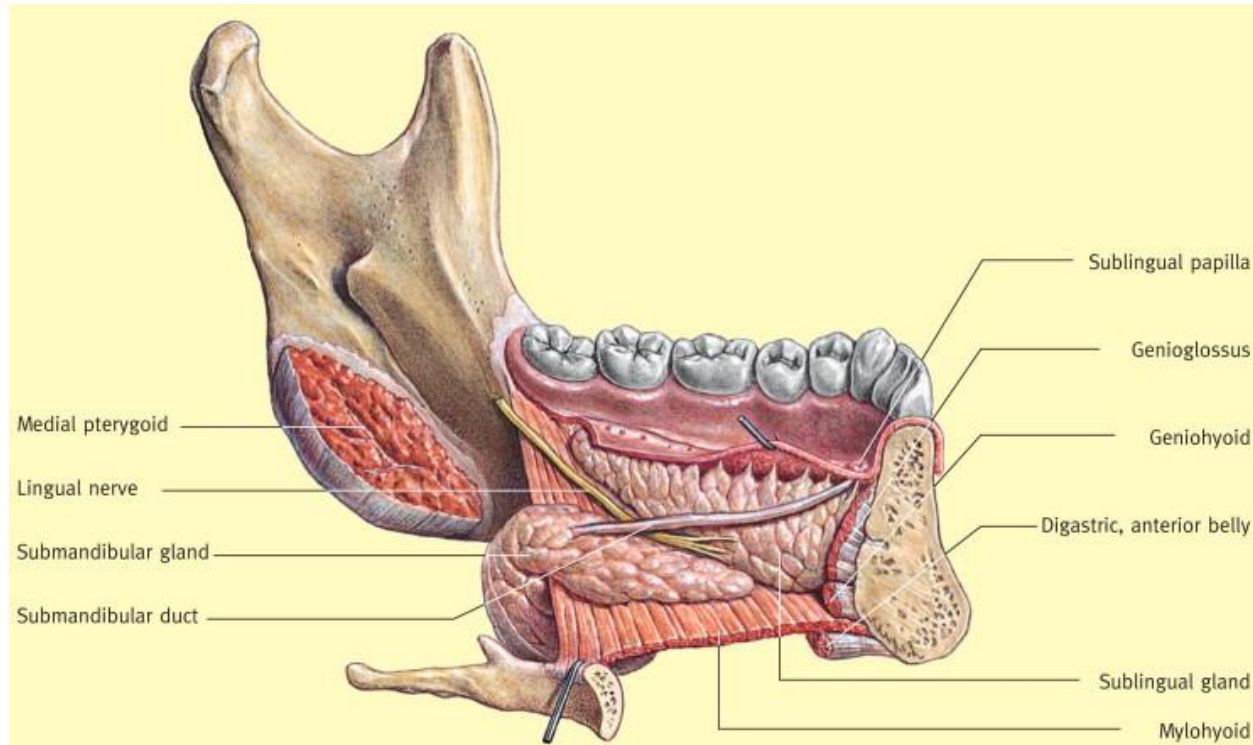
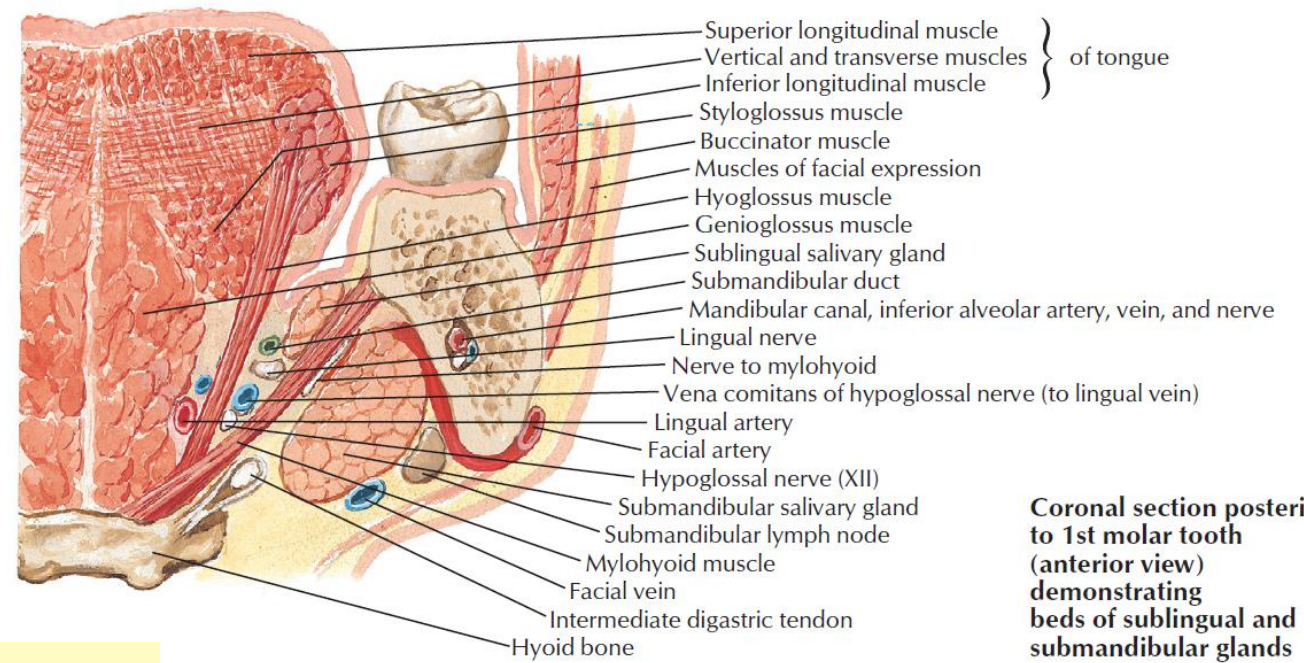
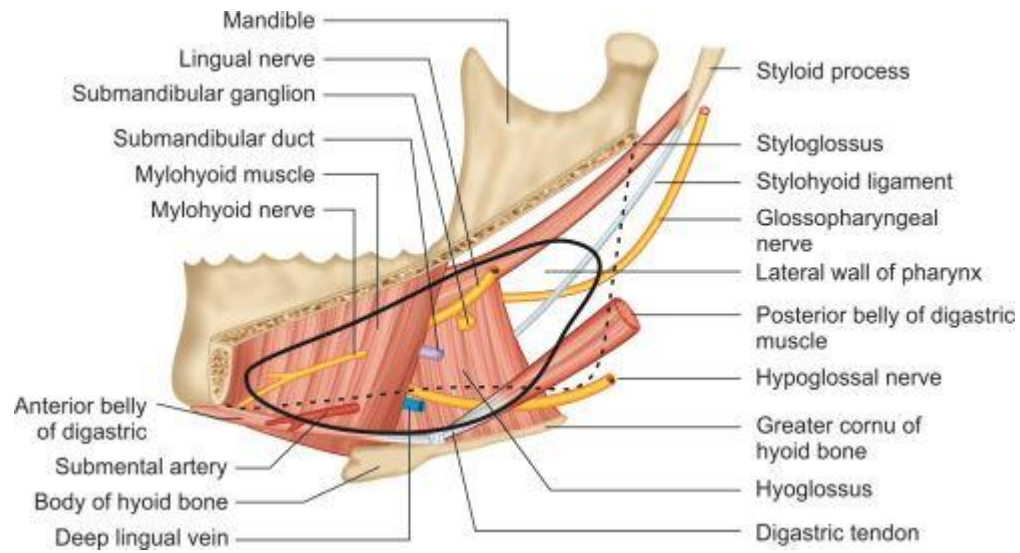
General Architecture: Mixed seromucous gland (predominantly serous) accounting for the majority of resting salivary secretion.

Location and Shape: Positioned in the submandibular triangle, draped across the posterior free edge of the mylohyoid muscle with a distinct U-shaped morphology.

Surfaces and Compartments:

Superficial Part: Rests in the submandibular triangle, bounded by skin, platysma, and deep cervical fascia.

Deep Part: Projects around the mylohyoid free border into the floor of the mouth.



Relations

5. Submandibular Gland: Relations & Ducts

Borders: Bounded superiorly by the mandible body and inferiorly by the digastric muscle bellies.

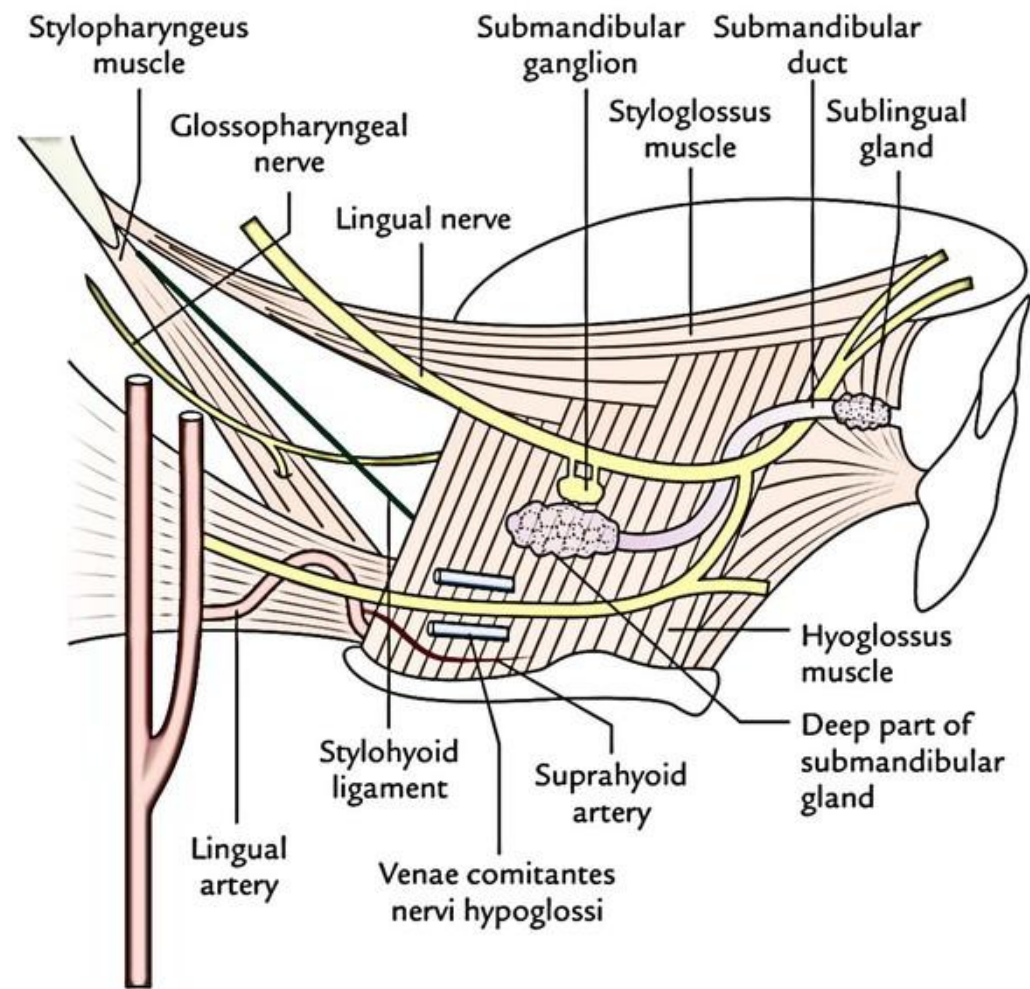
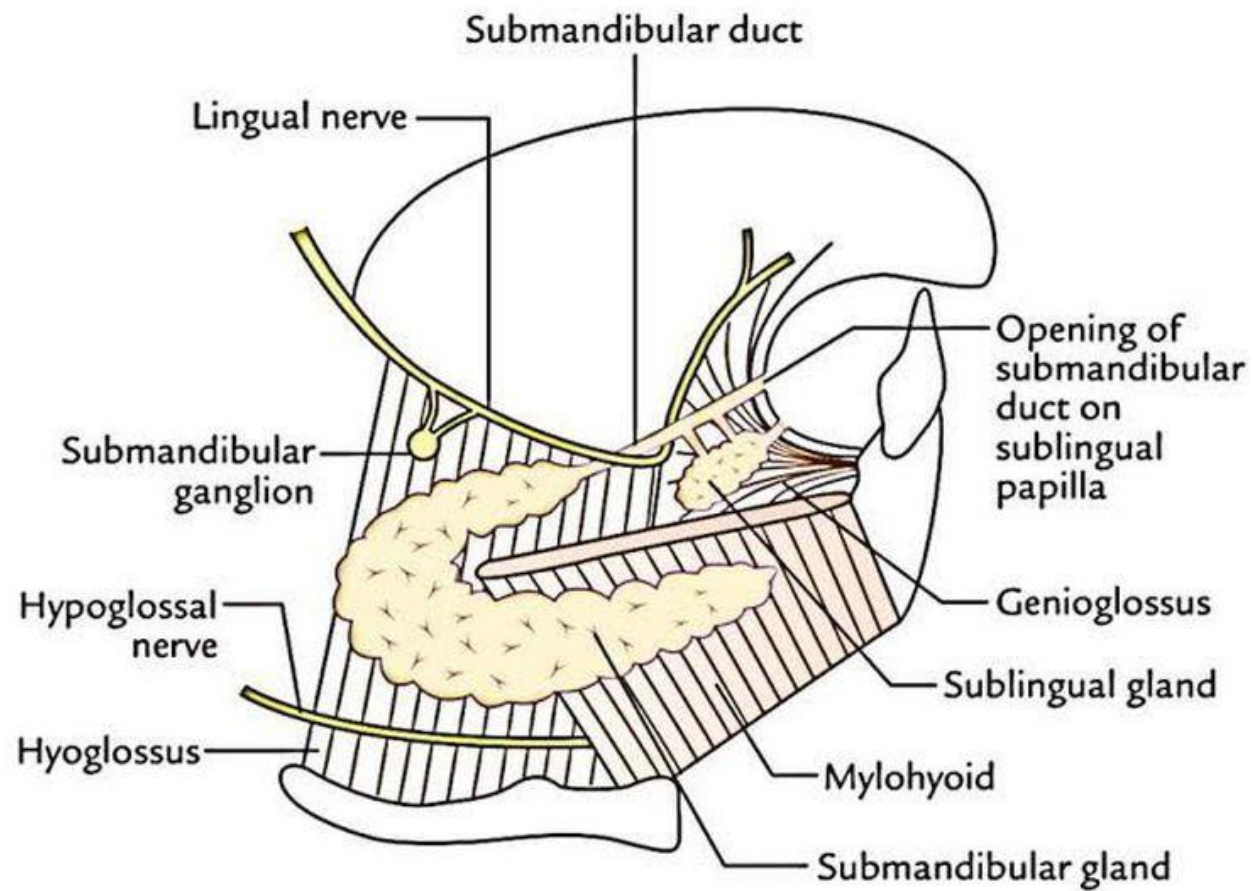
Neurovascular Relations:

Muscular Bed: Mylohyoid, hyoglossus, and middle constrictor muscles.

Facial Artery: Loops over the superior and posterior borders.

Nerves: Lingual and hypoglossal nerves pass deep; submandibular ganglion is suspended from the lingual nerve.

Ductal System (Wharton's Duct): Arises from the deep part, runs forward, crosses lateral to the lingual nerve, and opens onto the sublingual caruncle at the lingual frenulum base.



6. Sublingual Gland

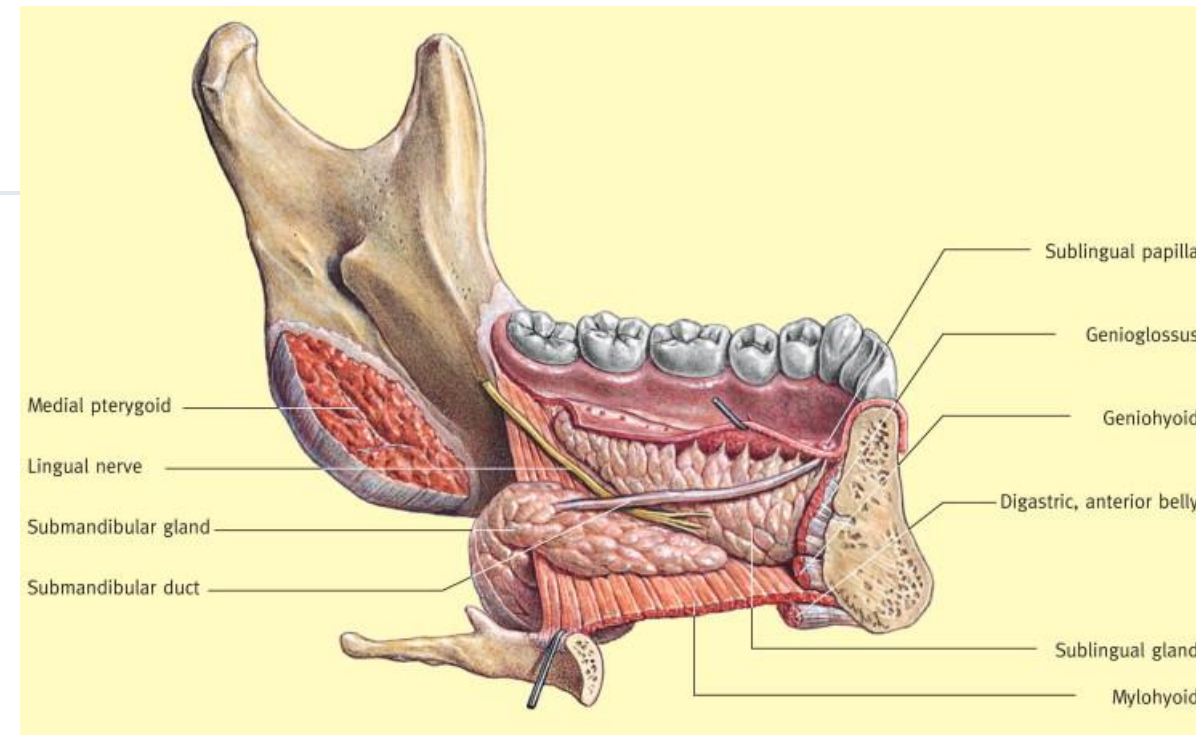
General Architecture: The smallest major salivary gland, primarily a mucous-secreting gland.

Lacks a true fibrous capsule.

Location and Surfaces: Lies directly beneath the floor-of-mouth mucosa, superior to the mylohyoid muscle. Elevates mucosa to form the sublingual fold.

Borders and Relations: Medially related to the genioglossus muscle and lingual nerve; laterally nestled against the sublingual fossa of the mandible.

Ductal System (Ducts of Rivinus): Drains via 8 to 20 independent minor ducts opening along the sublingual fold. Anterior ducts may join the submandibular duct to form Bartholin's duct.



7. Neurovascular Supply & Innervation of Salivary Glands

Arterial Supply: **Branches External Carotid Artery** (transverse facial, posterior auricular, facial, and lingual arteries).

Venous & Lymphatic Drainage: Parotid drains to retromandibular/jugular chains via parotid nodes; submandibular/sublingual drain to submandibular and deep cervical nodes.

Parasympathetic Innervation (Secretomotor):

1. **Parotid:** Inferior salivatory nucleus (CN IX) -> otic ganglion -> auriculotemporal nerve (CN V3).
2. **Submandibular & Sublingual:** Superior salivatory nucleus (CN VII) -> chorda tympani -> lingual nerve -> submandibular ganglion.
3. **Sympathetic Innervation:** Superior cervical ganglion via periarterial plexuses to reduce blood flow and produce thick, viscous secretions.



سوچا س